Chem 0010/11

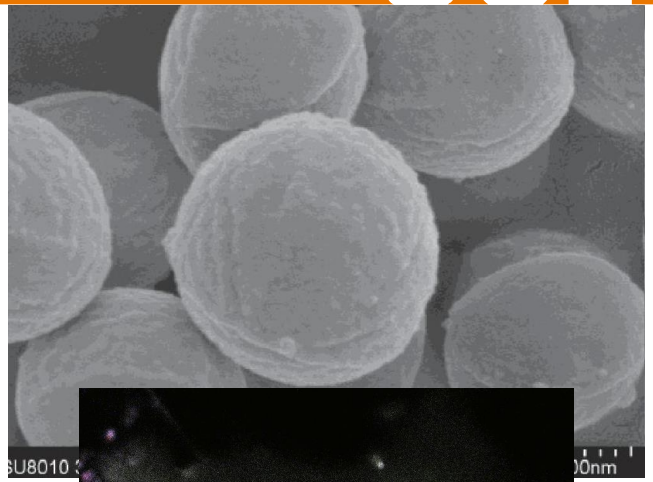
Light & its Properties

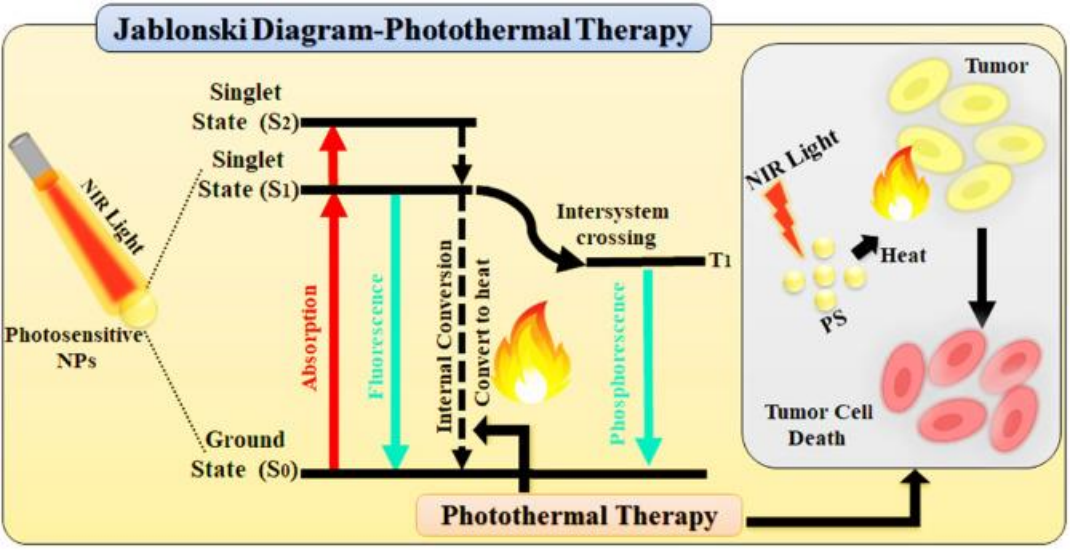
Dr Sam Cassidy
EngD, MRes, MSc, BSc, FHEA, MRSC, MInstP

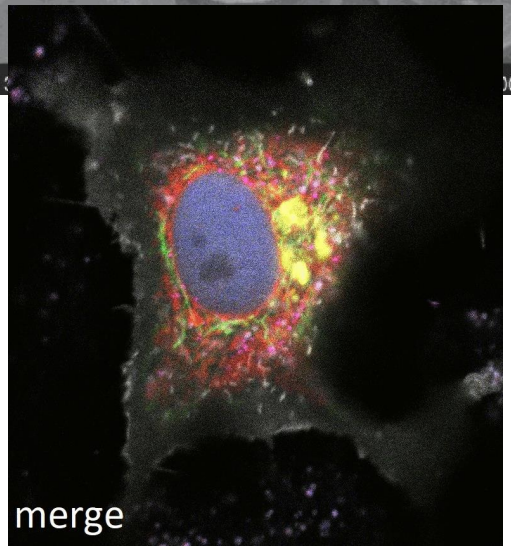


What is this section about?



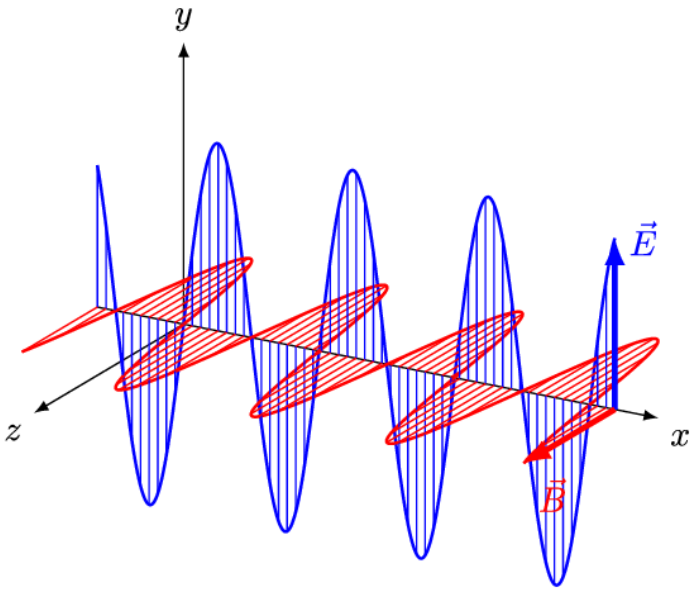






Classical Theory of Light

According to James Cleck Maxwell, Thomas Young, & Others, **light consists of oscillating electric and magnetic disturbances, perpendicular to each other, that propagate as waves.**



These waves move at the speed of light (c)

$$c = \frac{Distance\ of\ propagation\ during\ one\ period\ of\ oscillation}{Period\ of\ one\ oscillation}$$

c = 2.998… x 10^8 ***m/s*** in a vacuum = $\lambda\nu$

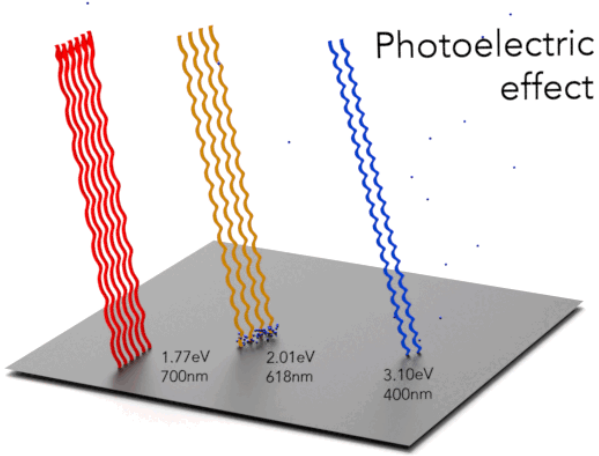
λ = Wavelength in ***m***

$\tilde{\nu}$ = Wavenumber in m^{-1} = $1/\lambda$

ν = Frequency in s^{-1} or ***Hz***

Quantum Theory of Light

According to Einstein, **light is made up of discrete packets of energy known as photons.** The photoelectric effect shows that light's ability to eject an electron is dependent on its frequency and not intensity.



Potassium - 2eV needed to eject electron

This was an extension on Plank's theory of quantised energy, where the energy of a photon $E = hv$.

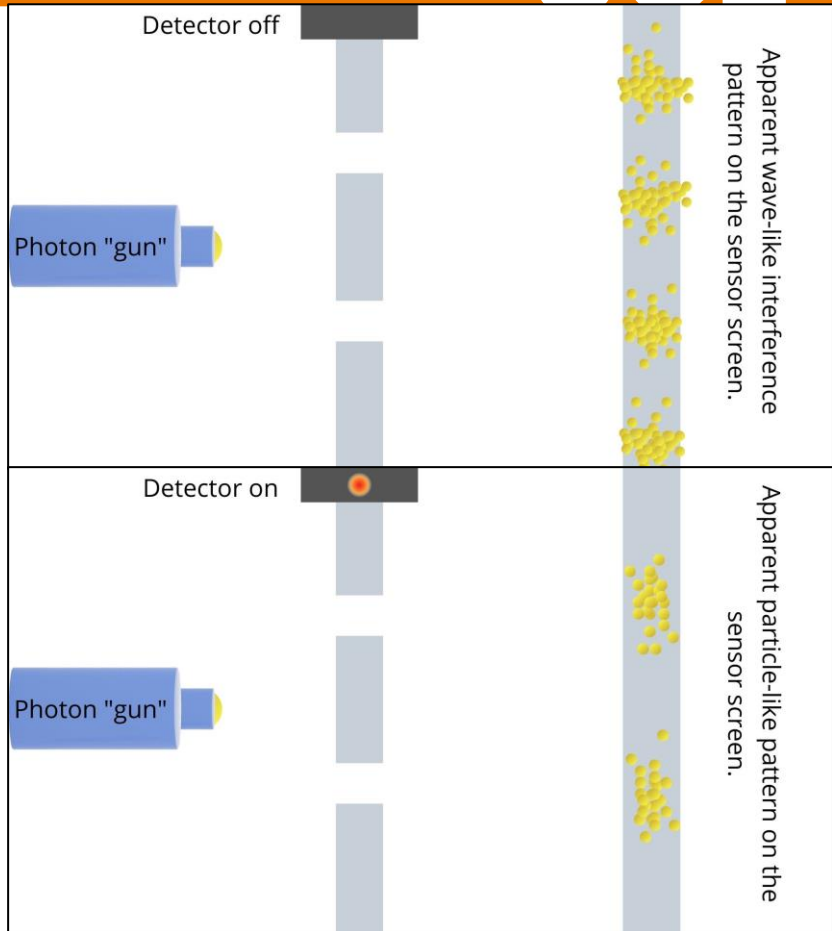
E = Energy of a photon in ***J***

h = Plank's constant 6.626×10^{-34} ***J s*** or ***J Hz^{-1}***

Which Theory is Correct

Young's double slit experiment can be used to show photons acting as both a wave and a particle.

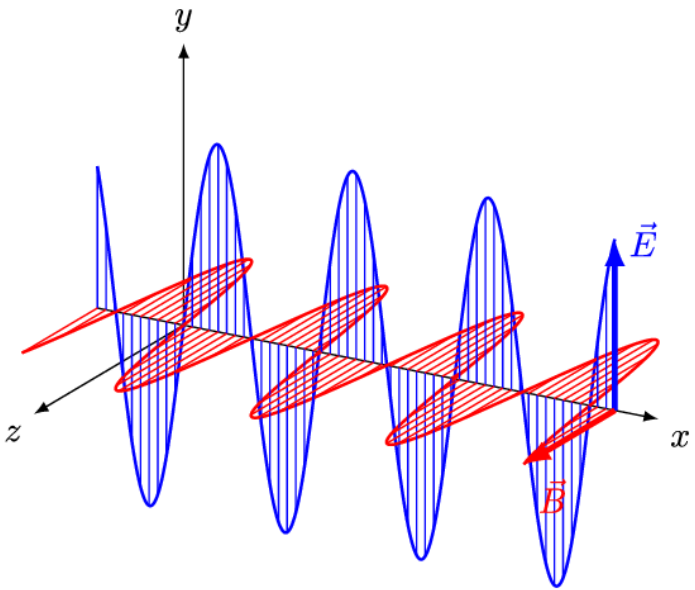
Louis de Broglie proposed that **not only does light act like a wave and a particle but so does everything else**.



Why Does This Matter

Neither the Classical **Theory of Light** or the **Quantum Theory of Light** can explain all the properties of light.

Wave-particle duality can.

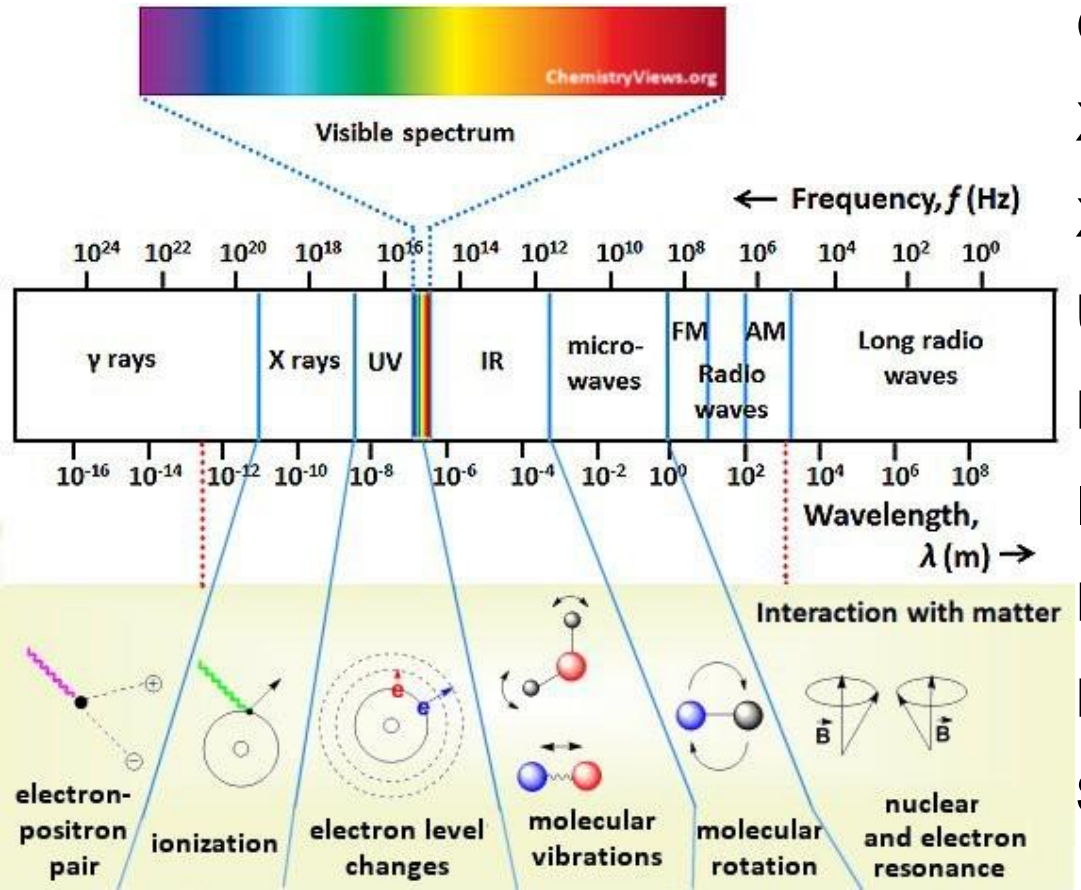


The wave properties of light are essential when we consider **diffraction**.

The particle properties of light are essential when we consider the **photoelectric effect**.

Both properties play important roles in **spectroscopy**.

The Electromagnetic Spectrum



Gamma-ray Spectroscopy

X-ray Diffraction Analysis

X-ray Photoelectron Spectroscopy

Ultraviolet-Visible Spectroscopy

Fluorescence Spectroscopy

Infrared Spectroscopy

Rotational Spectroscopy

Nuclear Magnetic Resonance Spectroscopy

How Light Interacts With Matter

Lab 14

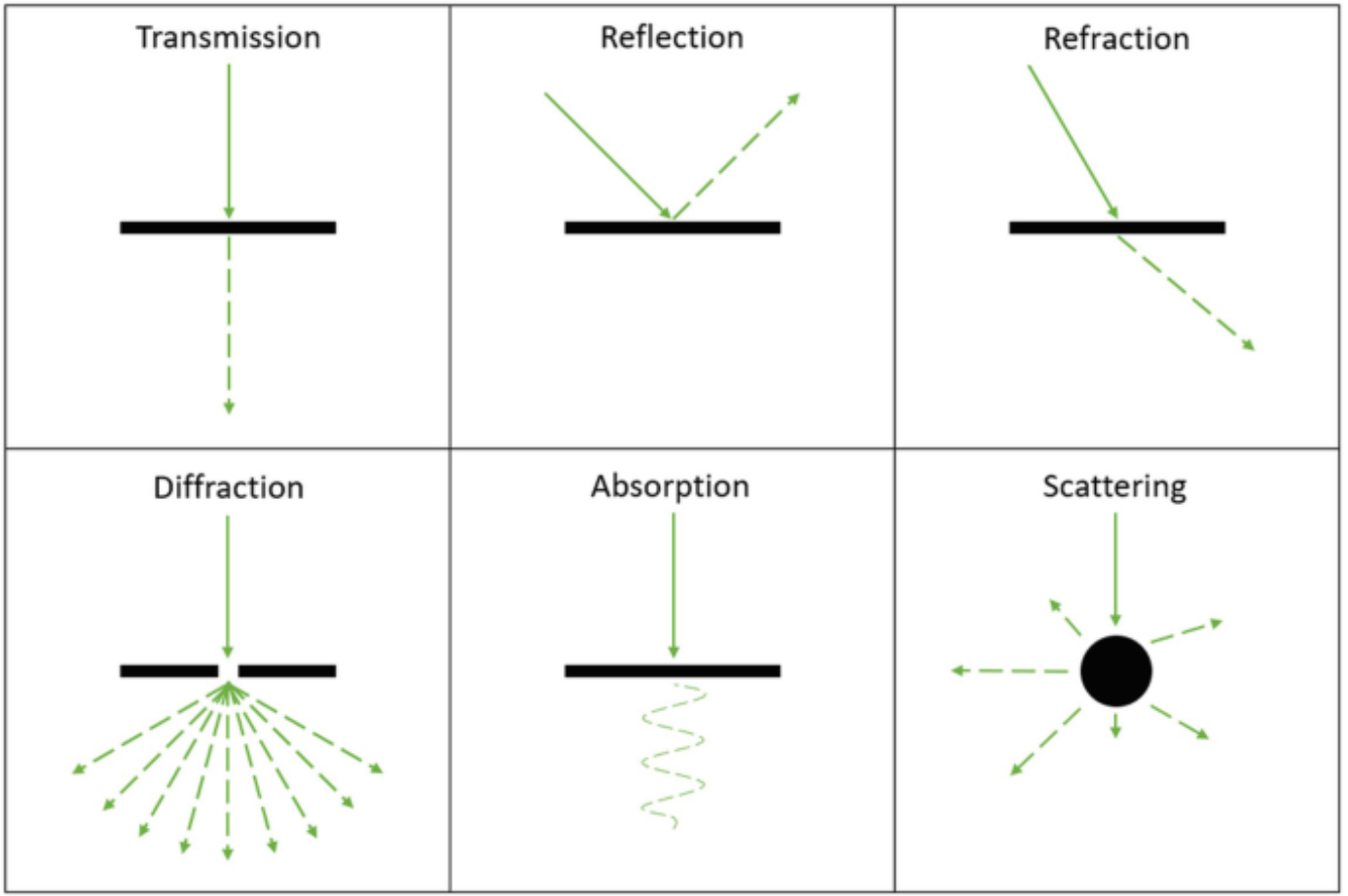
Chapter 6

Chapter 8

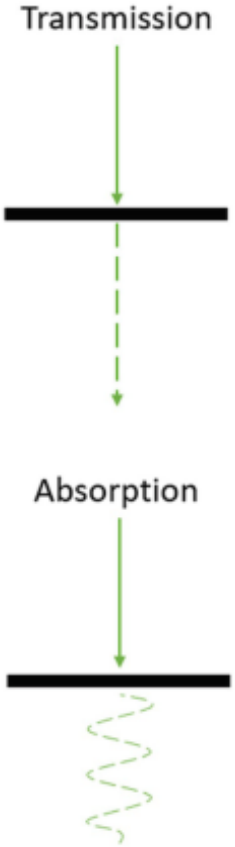
Chapter 17

and

Chapter 22



How Light Interacts With Matter



Transmission will not occur if the wavelength of light is a similar size to the particles (scattering) or if the wavelength of light has the correct energy to be absorbed.

Radio waves travel through walls because they are much larger than the particles that make up the wall.

Visible light travels through glass as it is not the correct energy to be absorbed.

Absorptions is an important part of spectroscopy and will be covered in detail in the next lecture.

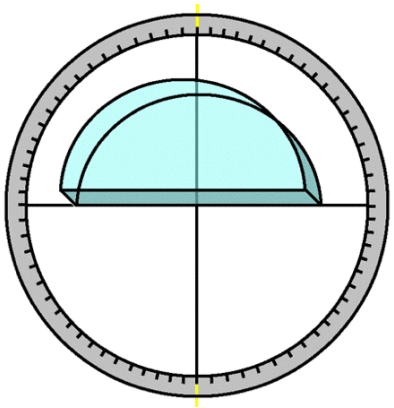
Refraction

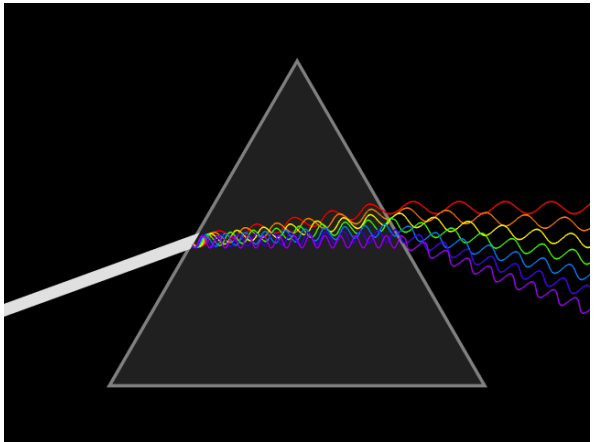
When light travels through a **vacuum**, **all light** travels 2.998 x 10^8 m/s (c).

When light is traveling through a medium, we need to consider the refractive index of the medium (n) where $n = c/v$ and $v = \lambda f$.

v = the speed of light in the medium ***m/s***

f = the frequency of light s^{-1}





Units can help differentiate between speed and frequency

Snell's law states $n_1 \sin\theta_1 = n_2 \sin\theta_2$

All light in a white light beam has the same n_1 and θ_1 from a vacuum.

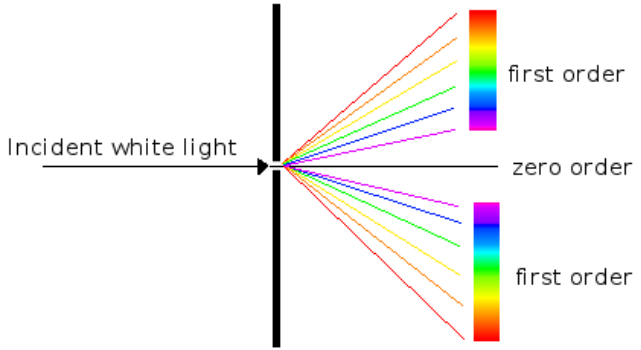
as n_2 **increases**, λ gets **shorter**

as λ gets **longer**, θ_2 **increases**

Diffraction

Diffraction is the bending of waves around an opening

The amount of bending is relative to the wavelength (λ)



$$d\sin\theta = n\lambda$$

d is the distance between slits

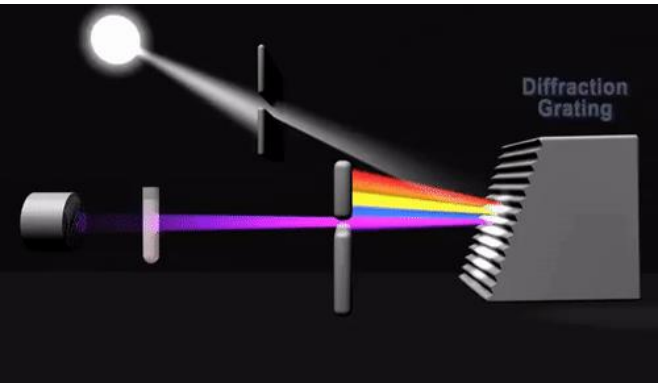
n is the order number of the maximum

θ is the diffraction angle

As λ **increases** so does θ

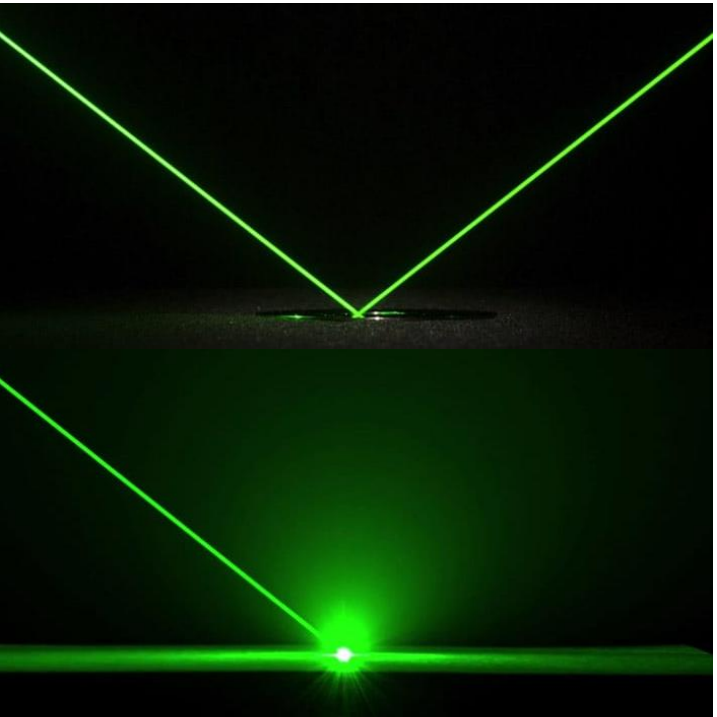
This is the **same** as **refraction**

This is how a spectrophotometer light source works



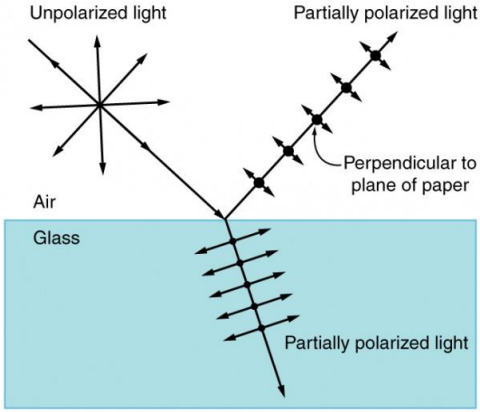
Reflection and Scattering

When light interacts with a surface that has free electrons (metals), it will not penetrate the surface and will instead be **reflected** or **absorbed**.



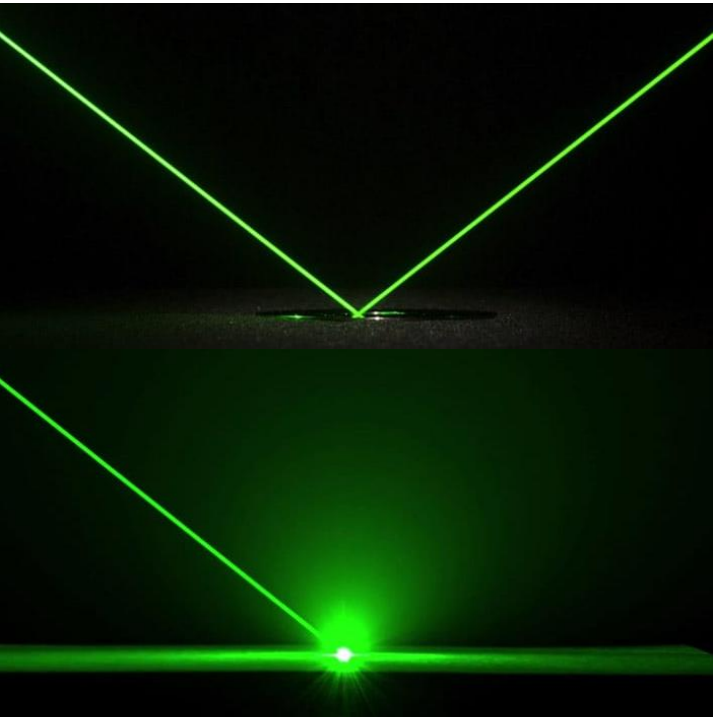
Dielectric surfaces (insulators) can also reflect light, but they can also undergo other interactions at the same time (refraction, scattering, etc.).

Lab 14 Chapters 17 and 22 you will look at how this interaction can polarise light



Reflection and Scattering

Scattering occurs when light interacts with particles that are on the same scale as the light's wavelength.



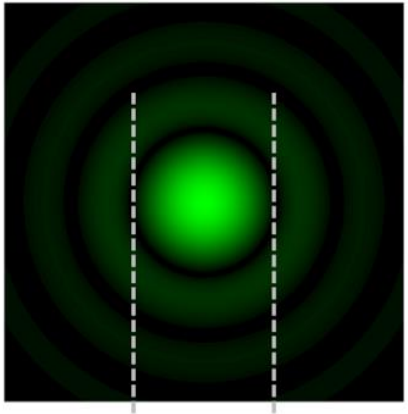
Rayleigh scattering is an elastic form of scattering from small particles.

Mie scattering is an elastic form of scattering from larger particles.

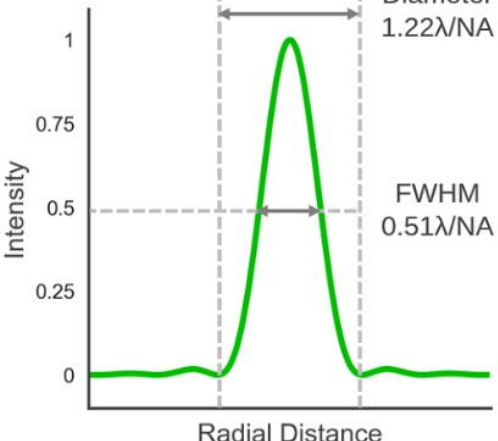
Diffuse reflection is the scattering that occurs when light interacts with a non-reflective surface.

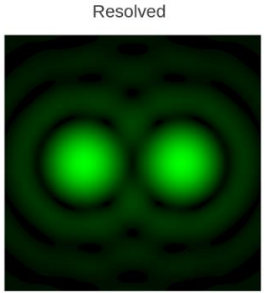
Non elastic scattering will be covered later.

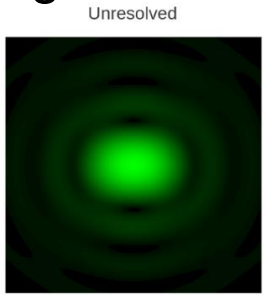
Microscopy



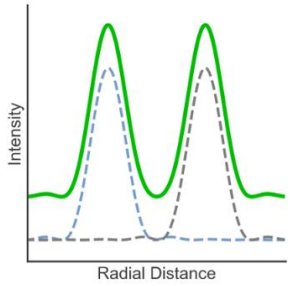
All microscopy is reliant on a source capable of illuminating a sample and a detector capable of observing the waves coming off a sample.

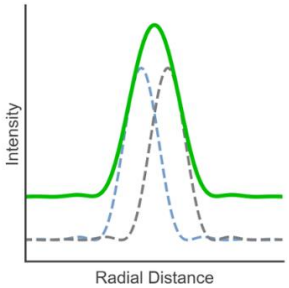






Diffraction limits the theoretical resolution of microscopes.

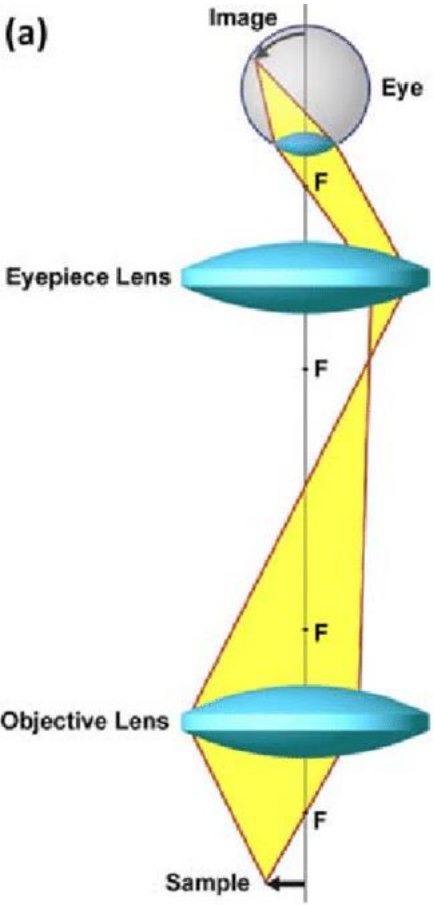




Airy Disc Diameter = 1.22λ/NA

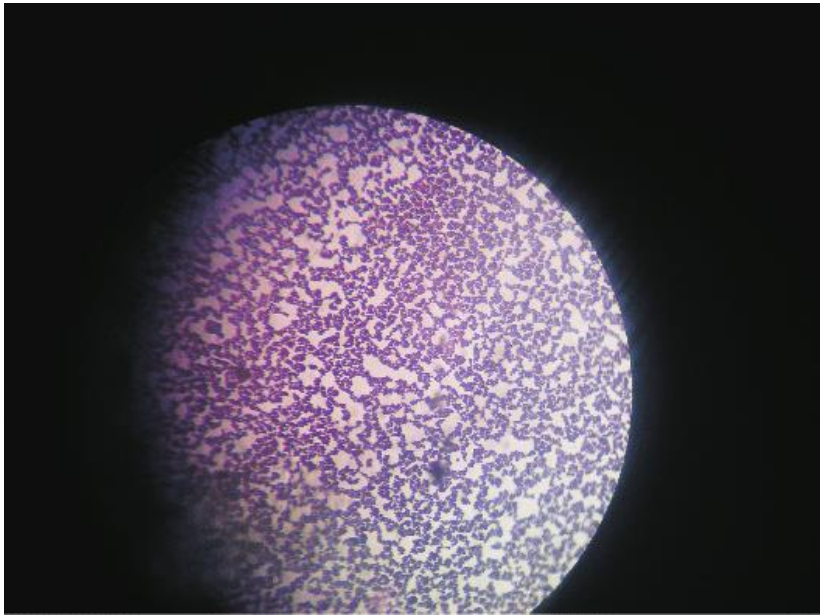
NA is the numerical aperture of the system (high NA ~1.4)

Microscopy



In **optical microscopy**, our eye is the detector, meaning it is limited to just visible light.

The best resolution for optical microscopes is ~200 nm

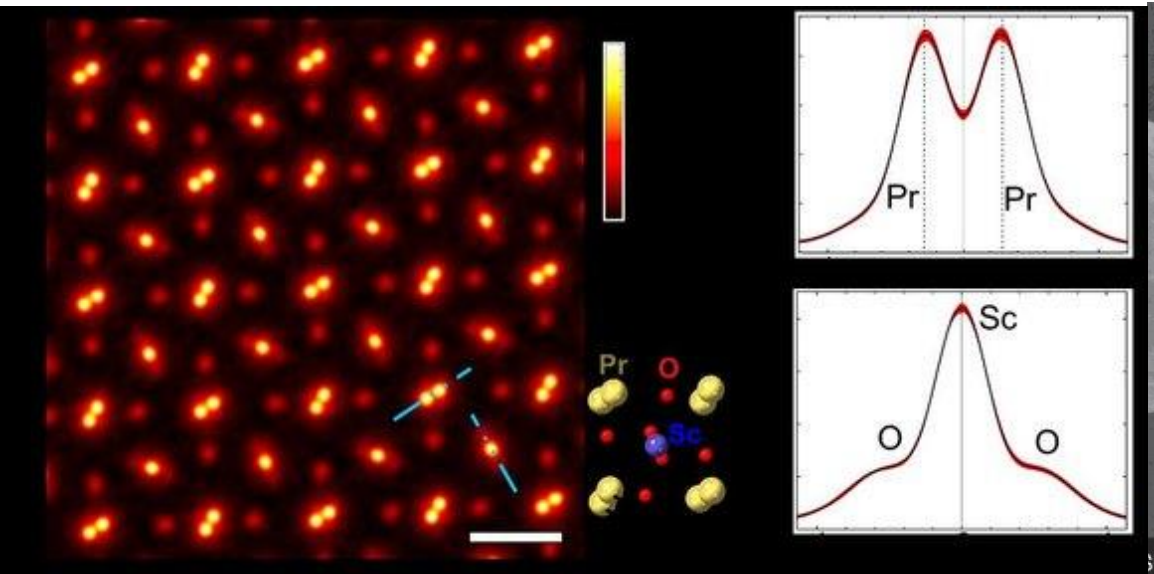


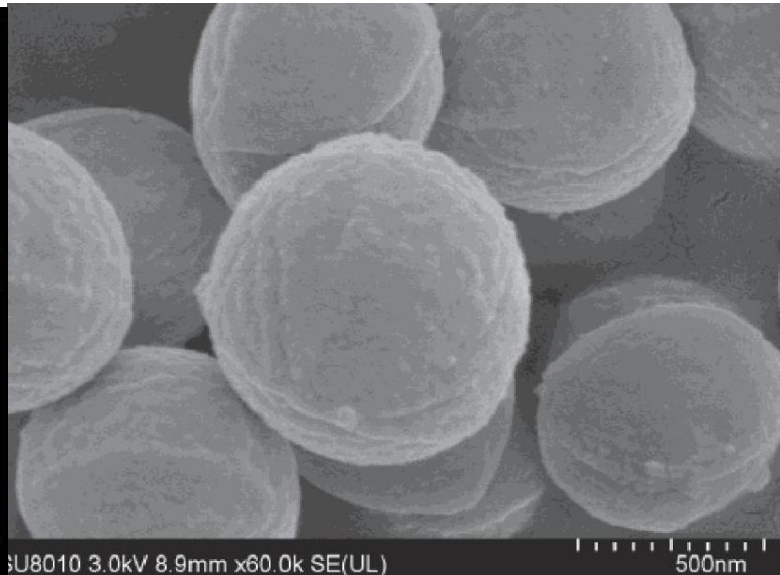
If everything is a wave, can we utilise particles with a smaller λ to increase our resolution?

Microscopy

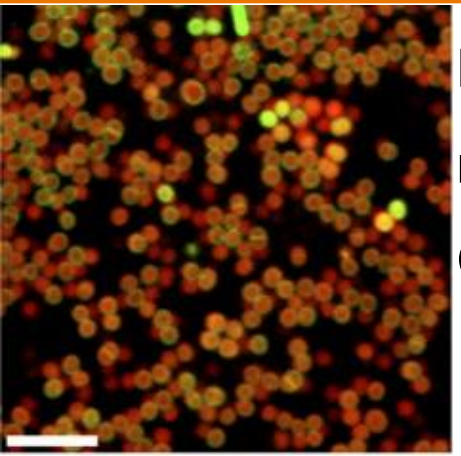
Electron microscopy makes use of <nm λ of electrons to achieve very high-resolution images.

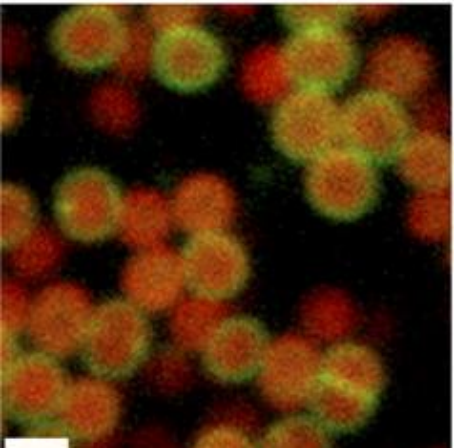
Generally, electron microscopy can only image inanimate/dead objects





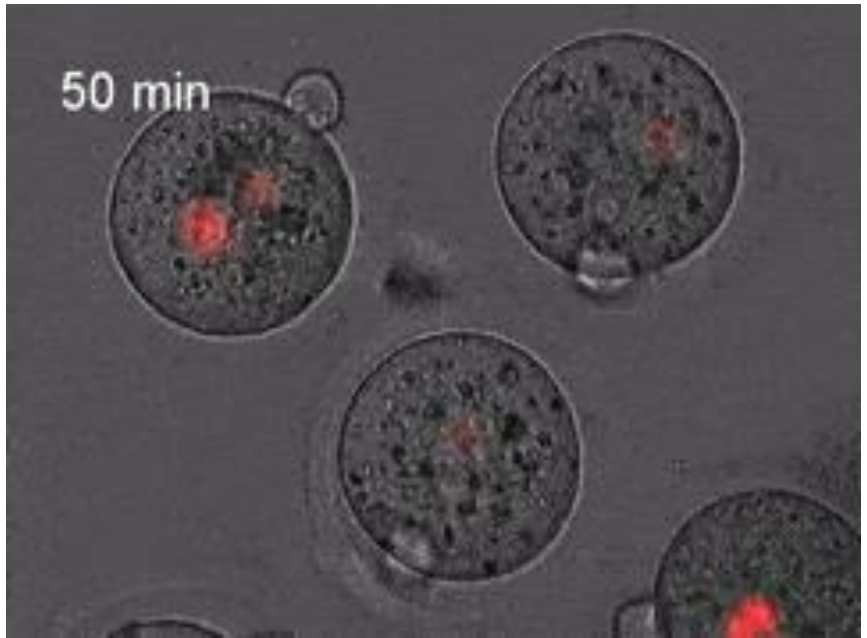
Microscopy

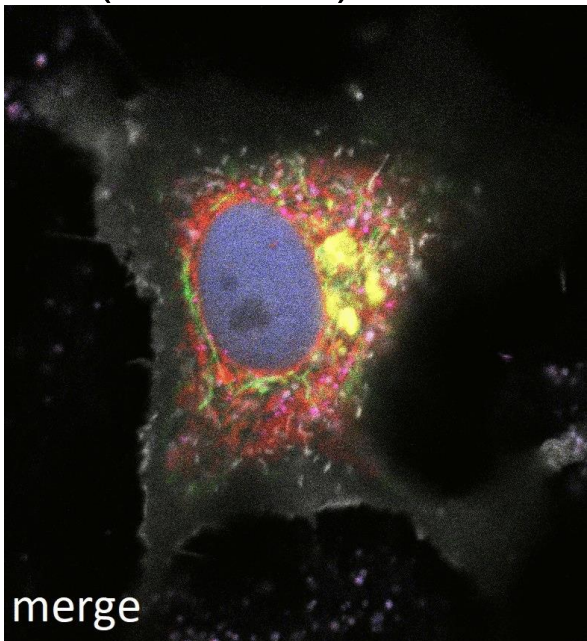




Light microscopy is excellent at imaging living cells, their movements, and interactions.

Can we go “Beyond the Diffraction Limit” (Lecture 3)





Summary

- Light is both a wave and a particle
- 6 ways light interacts with matter
- Calculate refraction and diffraction
- Different types of microscopy, and their limitations

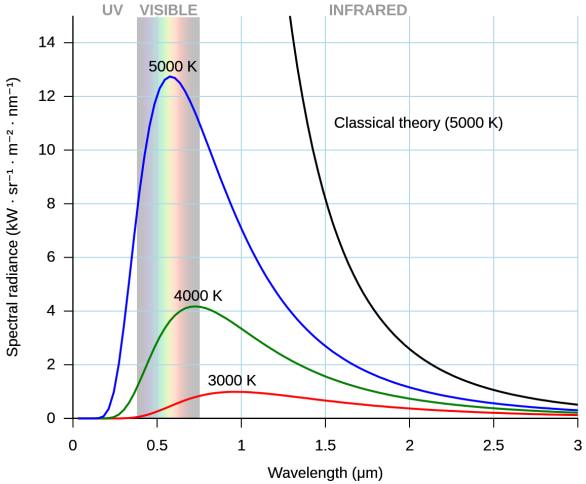
Chem 0010/11

Spectroscopy

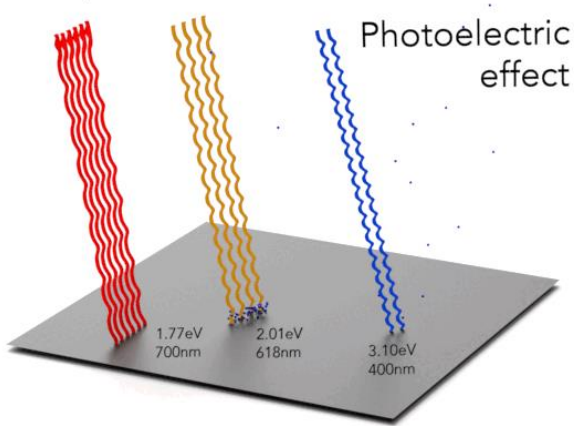
Dr Sam Cassidy
EngD, MRes, MSc, BSc, FHEA, MRSC, MInstP



Three Failures of Classical Physics

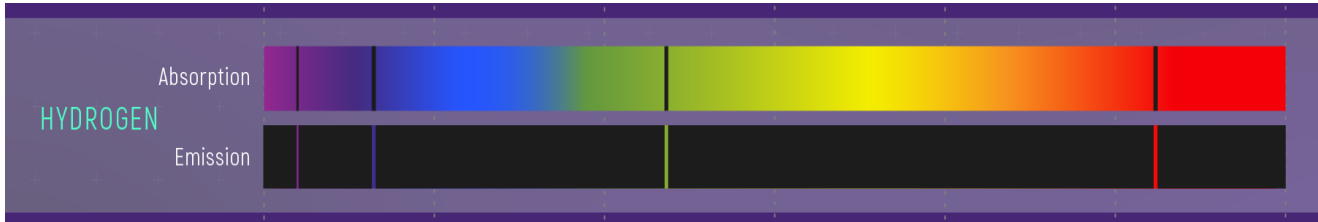


Black-Body Radiation (aka the Ultraviolet Catastrophe), Max Plank, 1900



The Photo-Electric Effect, Albert Einstein, 1905

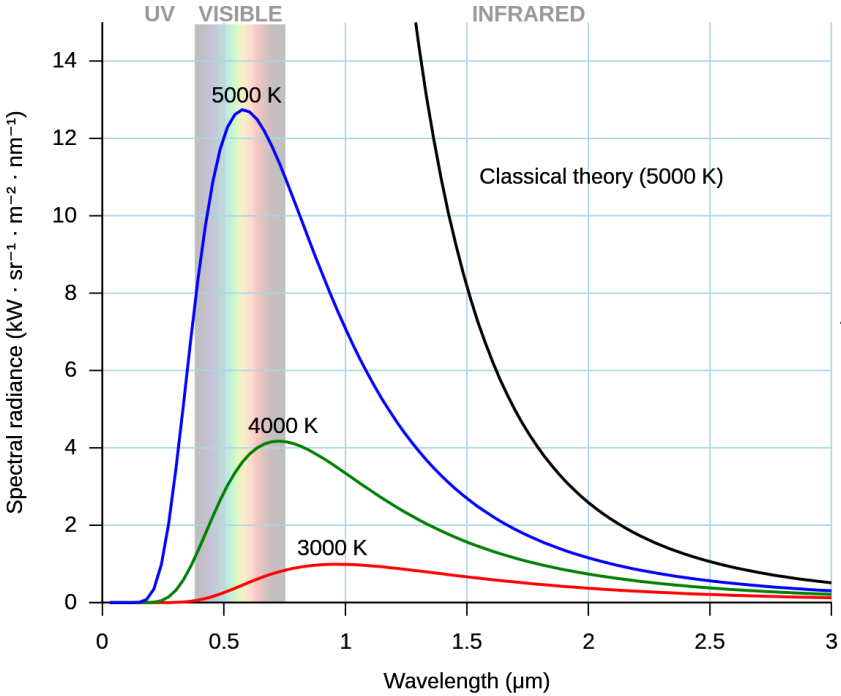
The Atomic Spectra, Niels Bohr, 1923



Black Body Radiation

A **Black Body** is a theoretical object that absorbs all light.

Stars and black holes are the closest things we have to black bodies.



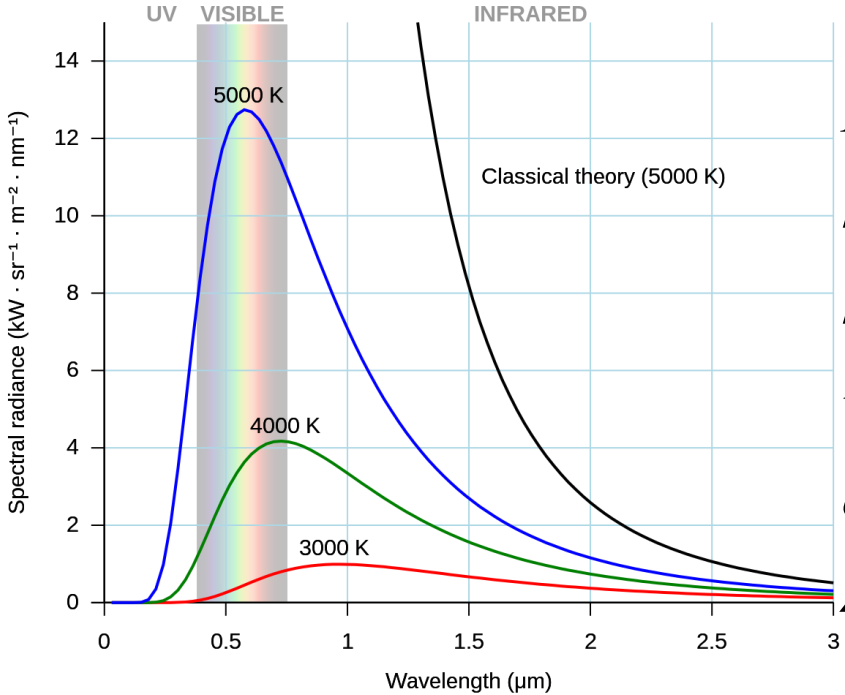
The black body spectrum depends on only the temperature of the object.

As temperature increases, the wavelength of the lambda max decreases.

The spectral radiance will not increase infinitely.

Black Body Radiation

To explain black body radiation Max Plank proposed the concept of **quantisation**.



$$E = nhv = nhc/\lambda$$

E = Energy ***J***

n = Integer 1, 2, 3, …

h = Plank's constant 6.626 x 10^{-34} ***J s***

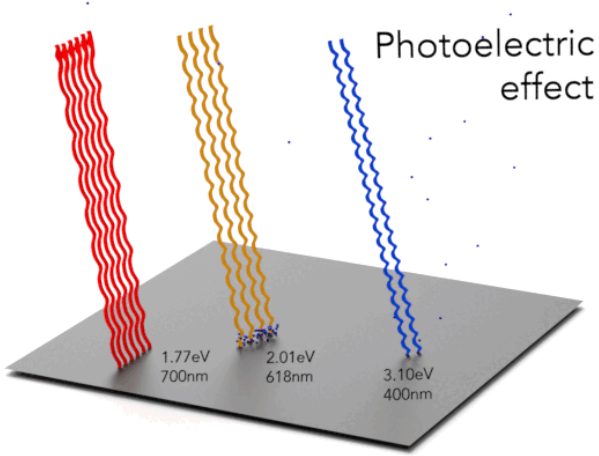
v = Frequency in s^{-1}

c = 2.998… x 10^{8} ***m/s*** in a vacuum

λ = Wavelength in ***m*** or ***nm***

The Photoelectric Effect

According to Einstein, **light is made up of discrete packets of energy known as photons.** The photoelectric effect shows that light's ability to eject an electron is dependent on its frequency and not its intensity.



Potassium - 2eV needed to eject electron

This was an extension on Plank's theory of quantised energy, where **light is also made of quantised.**

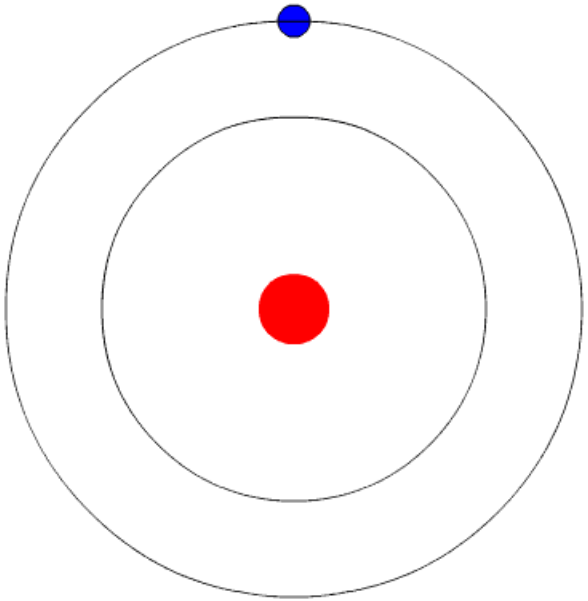
$$E = hv$$

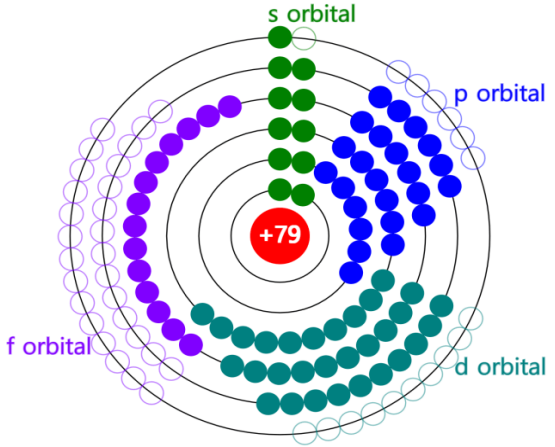
E = Energy of a photon in ***J***

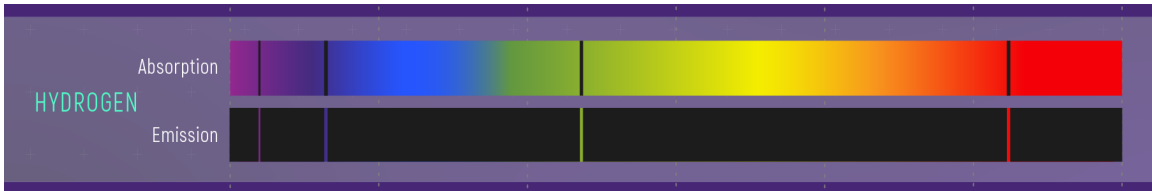
h = Plank's constant 6.626×10^{-34} ***J s*** or ***J Hz^{-1}***

The Bohr Model

Niels Bohr proposed that **the potential energy of an electron in a hydrogen atom is also quantised**. This means that electrons inhabit certain energy levels that are a fixed distance from the nucleus.

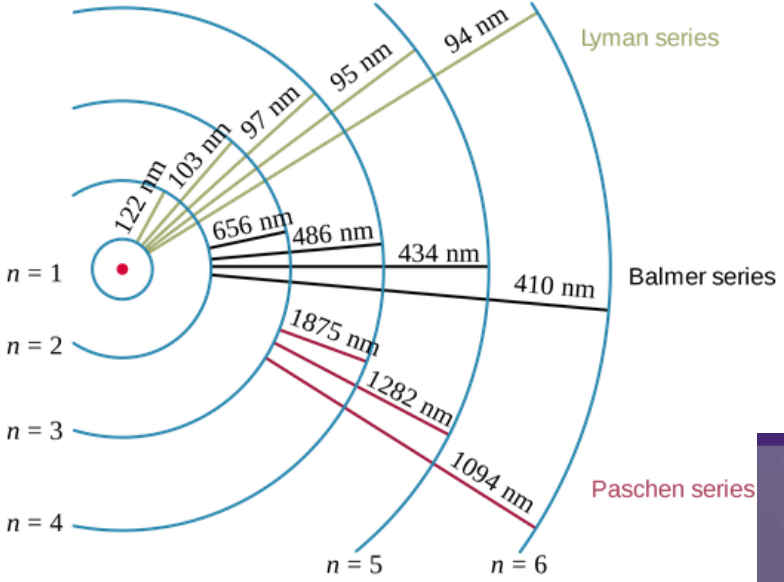






Atomic Energy Levels

As you have seen in ***Section A***, electrons orbiting an atom are in shells, with each electron's shell defined by the **principal quantum number "*n*"**, where ***n*** = Integer 1, 2, 3, …



$$\Delta E = h\nu = hc/\lambda$$

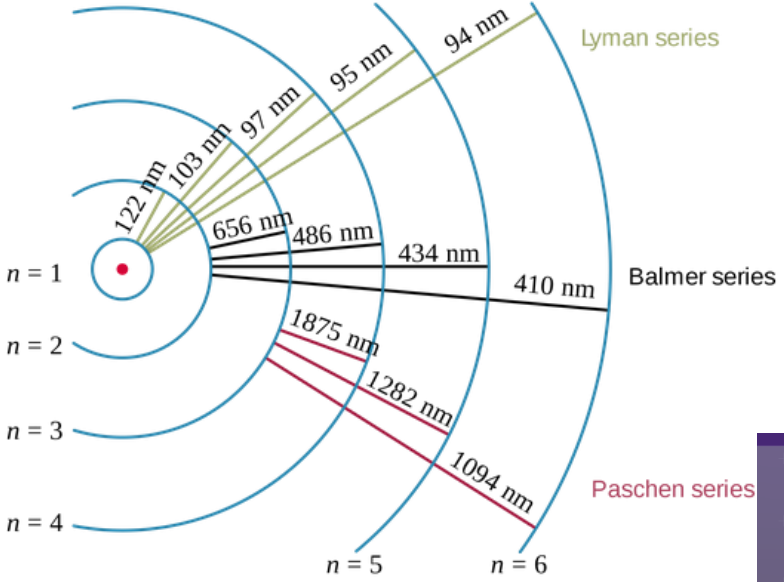
ΔE = is the change in energy ***J***

ΔE represents the difference in energy between the orbitals an electron is transitioning between.



Atomic Spectroscopy

Johann Balmer first presented an empirical formula capable of determining the wavelengths of spectral lines. However, this was specific to some hydrogen transition.



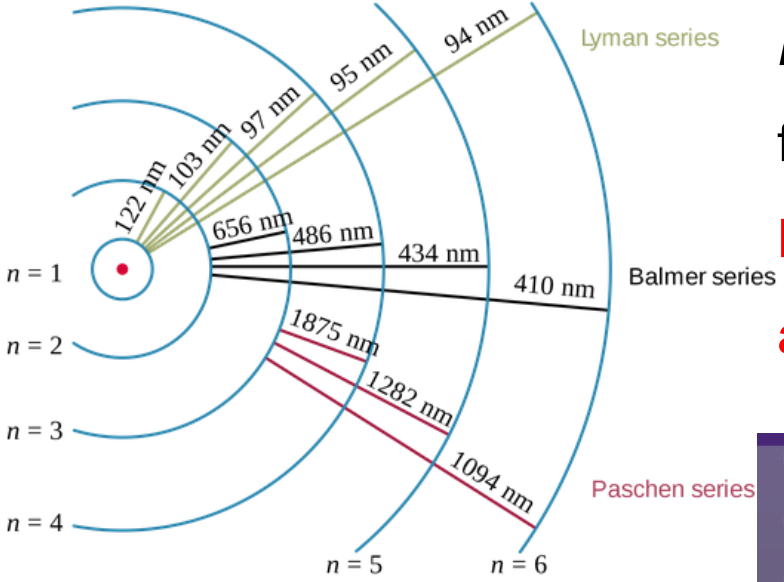
Fortunately, this was later generalised by Johannes Rydberg for all atomic electron transitions.



The Rydberg Equation

$$\tilde{v} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$R_H = 109677.8\ cm^{-1}$$



$\tilde{\nu}$ = Wavenumber

Z = the charge of the nucleus

R_H = the Rydberg constant for Hydrogen

n_1, n_2 = integers representing the initial and final energy levels (n2 > n1)

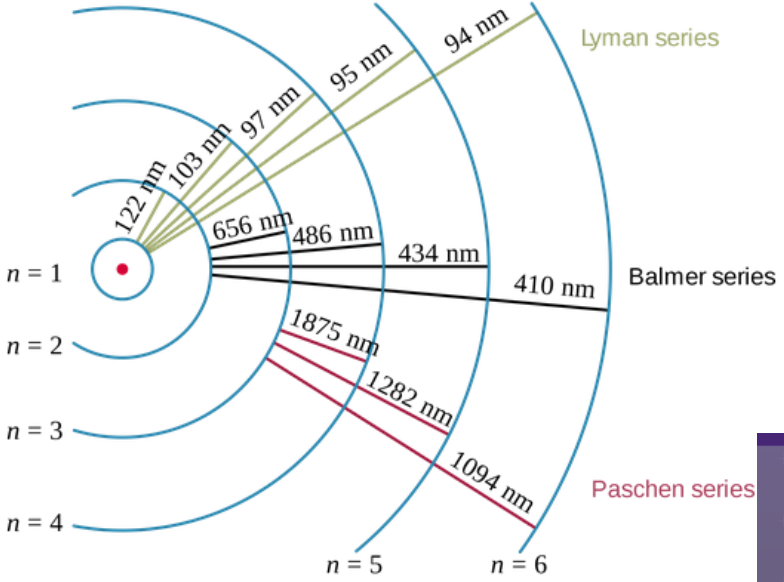
Becomes more complex when dealing with a multi electron system.



The Rydberg Equation

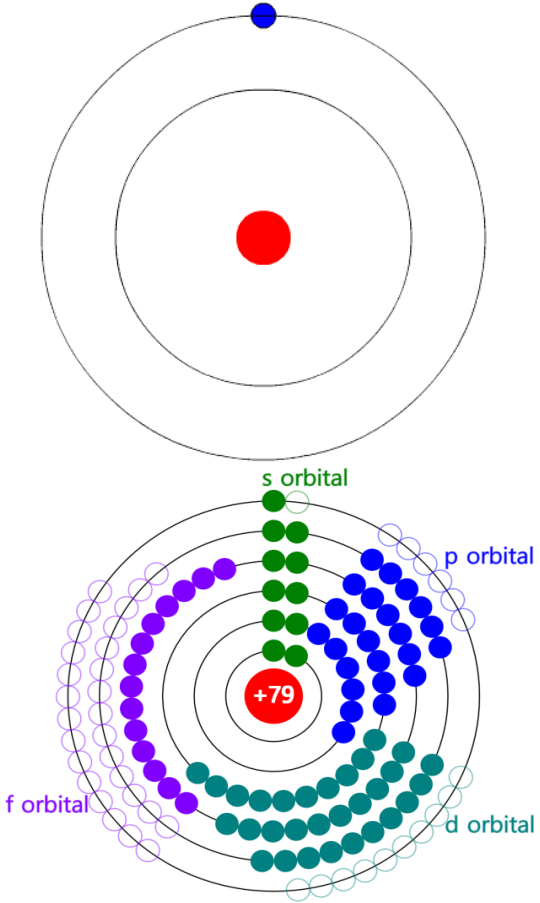
$$\tilde{v} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

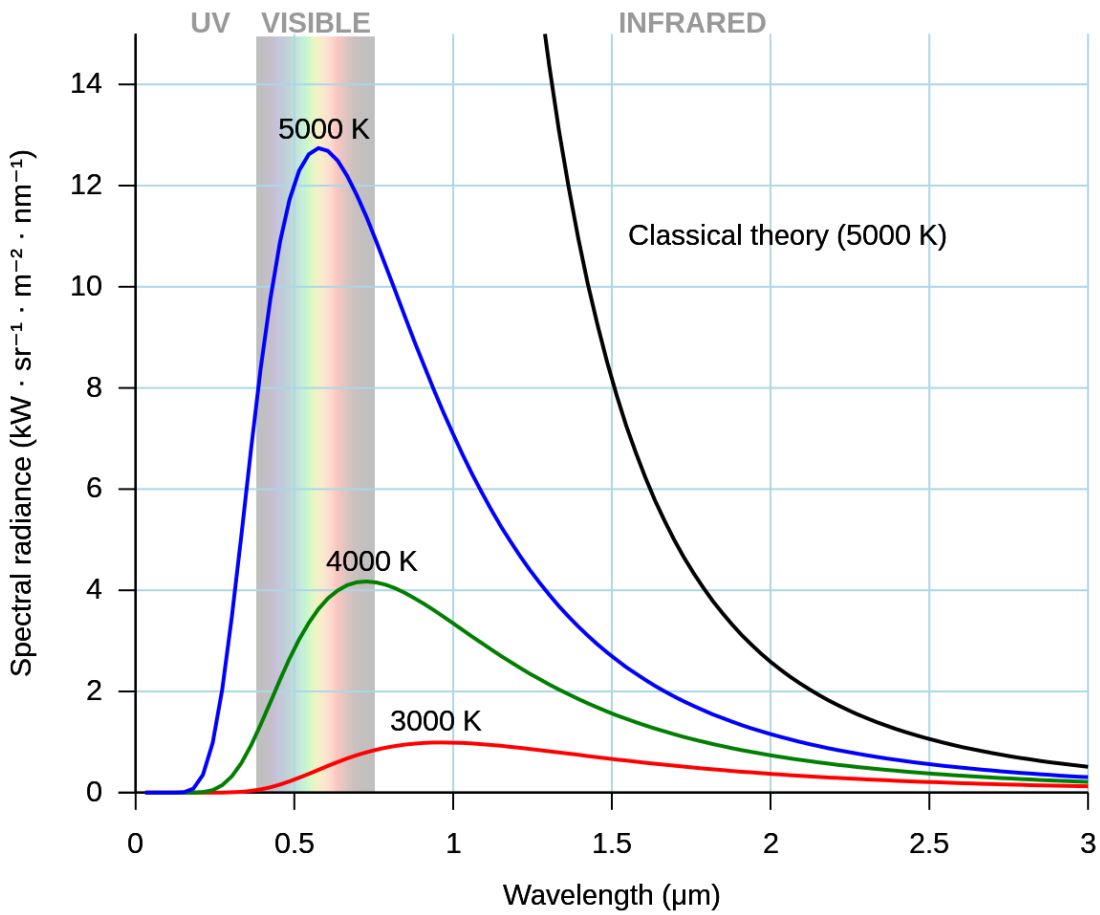
$$R_H = 109677.8\ cm^{-1}$$

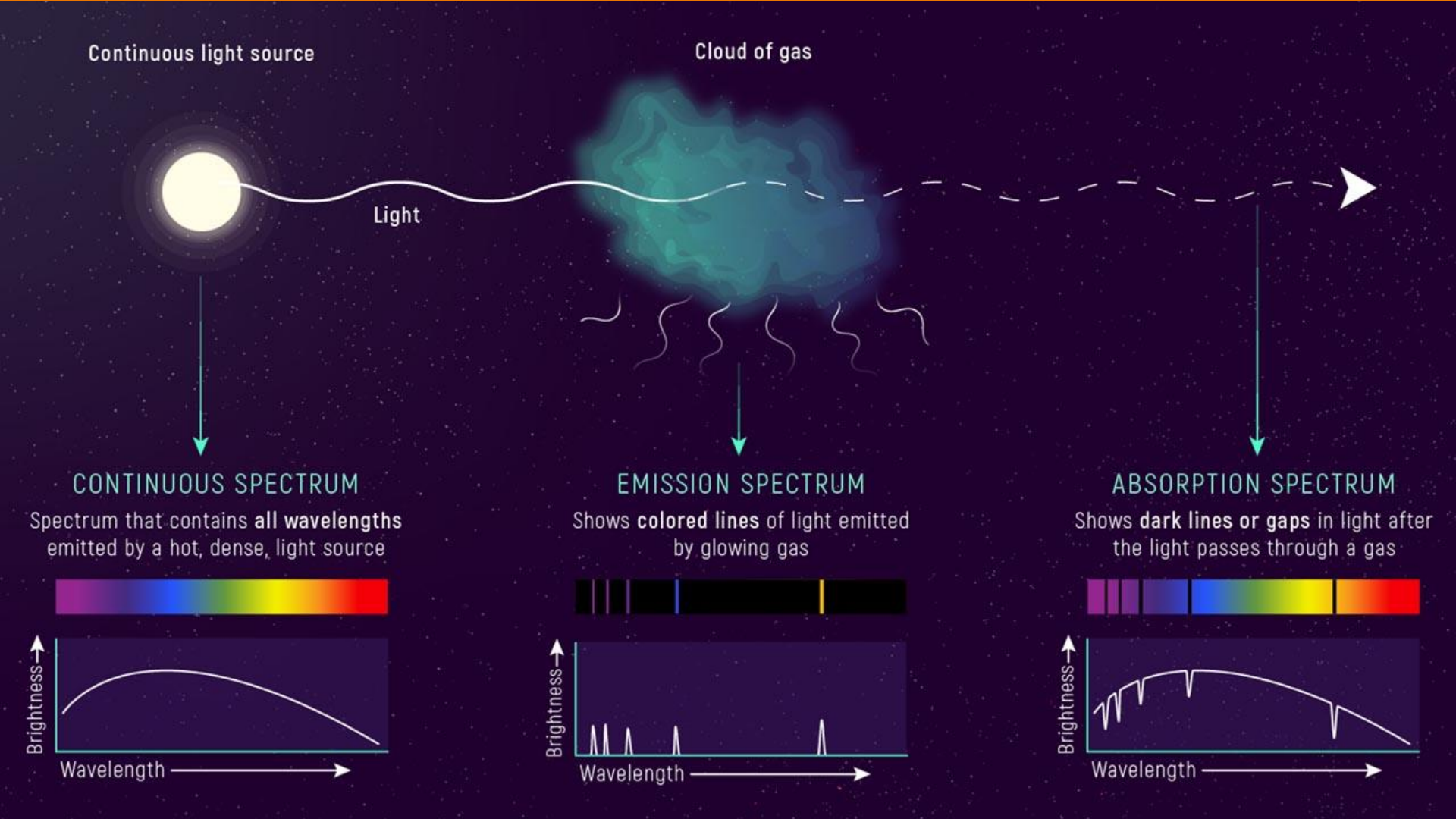




How a Black Body Radiates

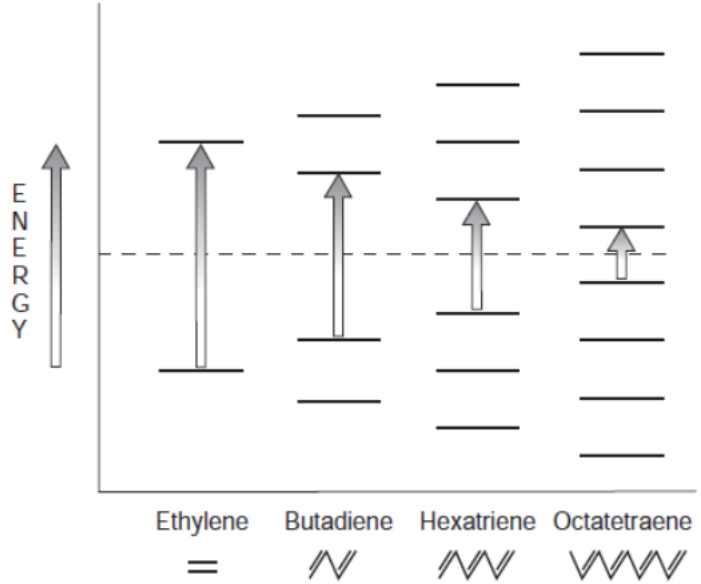





Continuous light source
Cloud of gas
Light
CONTINUOUS SPECTRUM
Spectrum that contains all wavelengths emitted by a hot, dense, light source
Brightness
Wavelength
EMISSION SPECTRUM
Shows colored lines of light emitted by glowing gas
Brightness
Wavelength
ABSORPTION SPECTRUM
Shows dark lines or gaps in light after the light passes through a gas
Brightness
Wavelength

Molecular Energy Levels

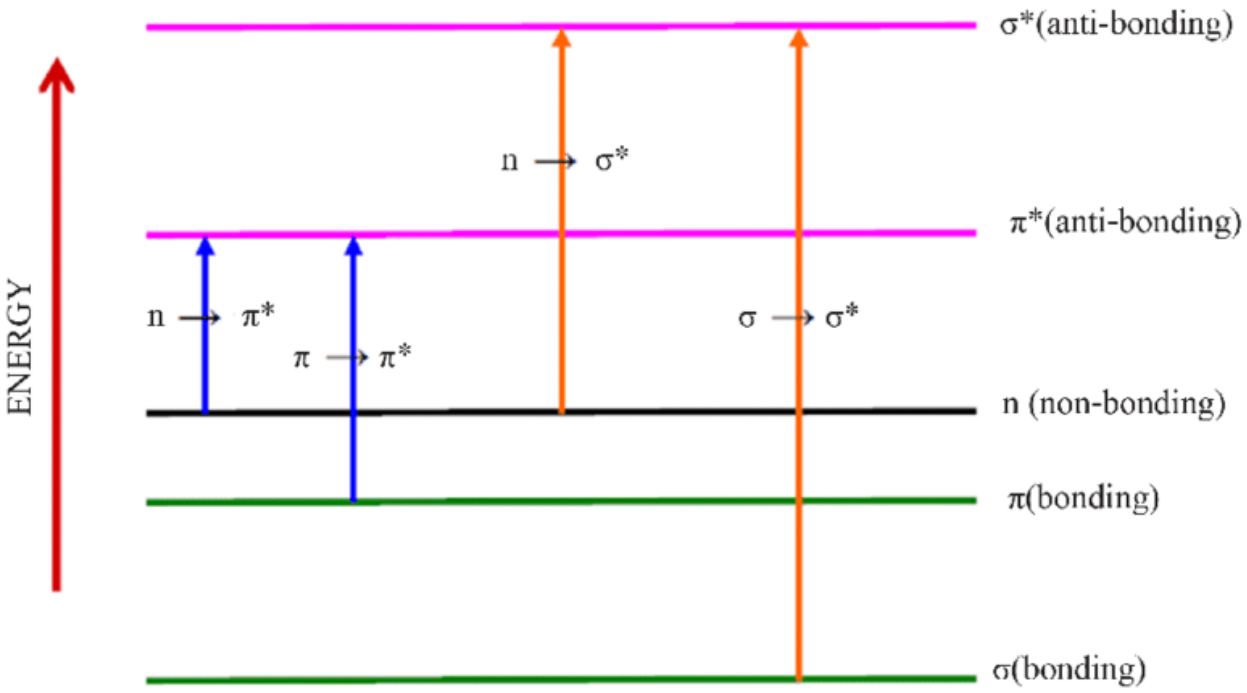
When looking at **Polyunsaturated Hydrocarbons** molecules the lowest energy transition will be the $\pi - \pi^*$ transition. This means the **band gap** (HOMO-LUMO, eV) will be reduced with the addition of each π bond.

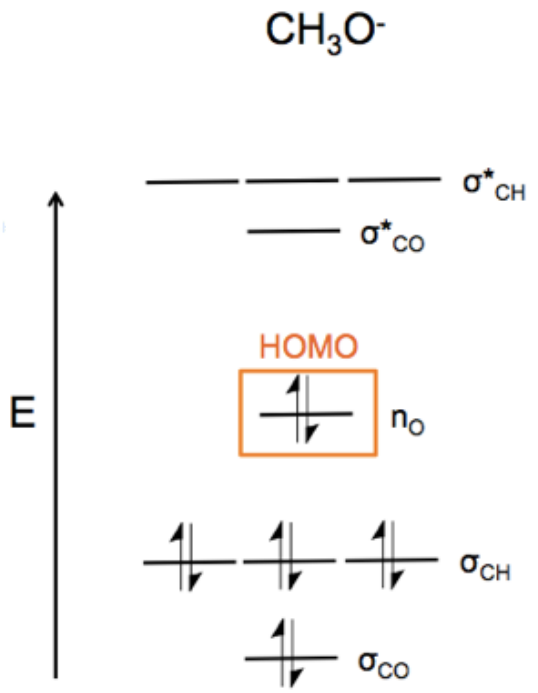


Compound	λ_{max} (nm)
$H_2C{=}CH_2$	165
	217
	256
	290
	334
	364

Molecular Energy Transitions

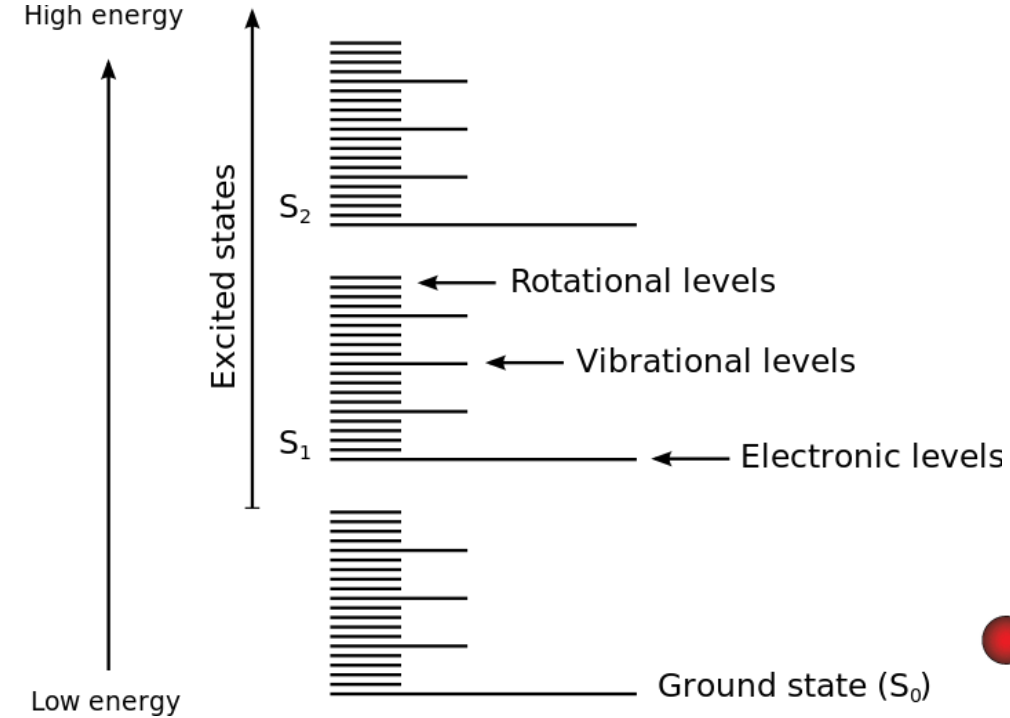
The addition of a bond may not always impact band gap of a molecule.

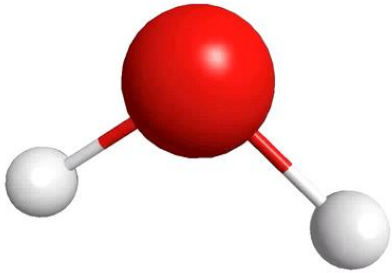


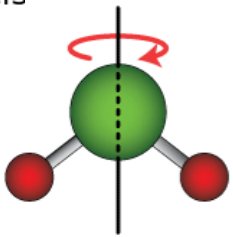


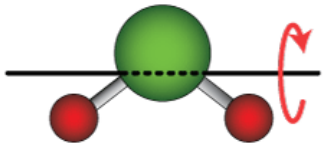
Molecular Spectroscopy

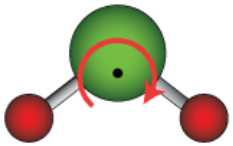
In molecular spectroscopy, energy interactions can result in vibrational and rotational transitions, not just electronic transitions.











Quantitative Spectroscopy

In spectroscopy, the ***Beer-Lambert Law*** can be used to relate concertation to transmission, and in turn absorption.

$$A = \varepsilon lc,\ T = I/I_0,\ A = -\log T$$

A = absorbance

ε = molar absorption coefficient $Lmol^{-1}cm^{-1}$

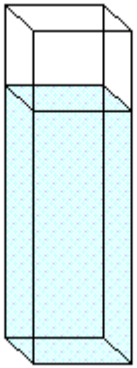
l = path length ***cm***

c = concentration $molL^{-1}$

T = transmittance

I = intensity of light reaching the detector

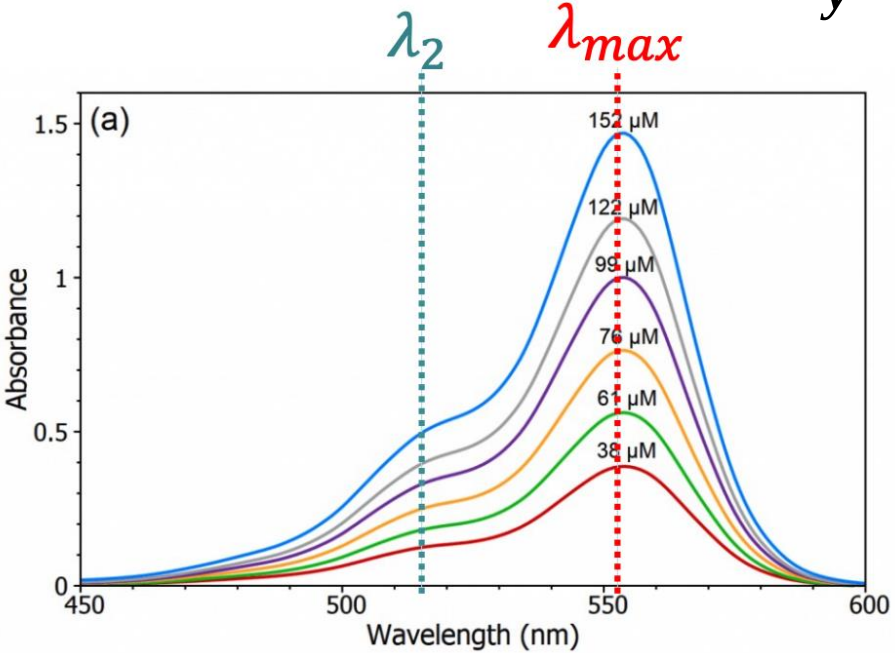
I_0 = intensity of light emitted by the source

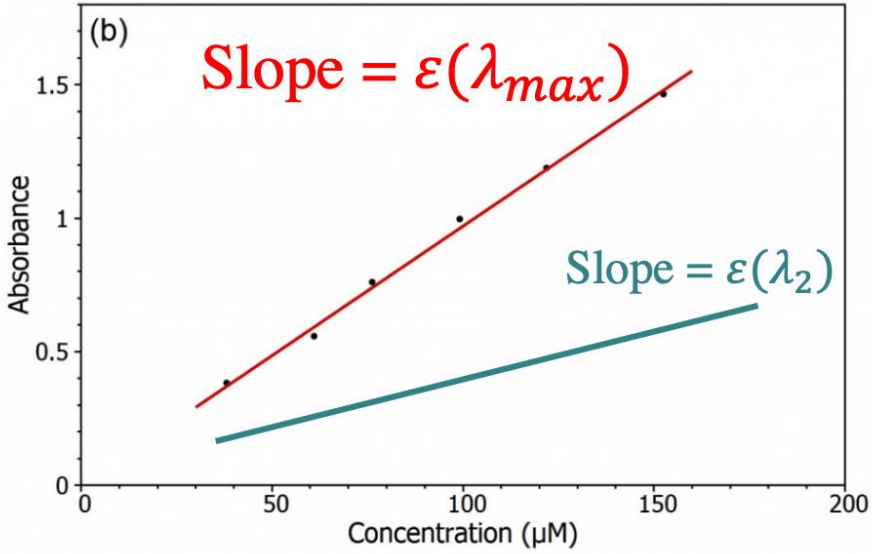


Quantitative Spectroscopy

To determine ε (specific to a substance and λ), a standard curve of A (y) vs c (x) is plotted, with the value of the slope (m) representing ε x l.

$$y = mx + c$$





Quantitative Spectroscopy

A solution is found to have a ε of 1.19897×10^5 $Lmol^{-1}cm^{-1}$, with an unknown concertation registering a transmission value of 42%. l is 0.01 m, determine the concentration.

Pulse Oximetry



Summary

- The three failures of classical physics
- Atomic spectroscopy
- Molecular spectroscopy
- The Rydberg Equation
- Quantitative spectroscopy

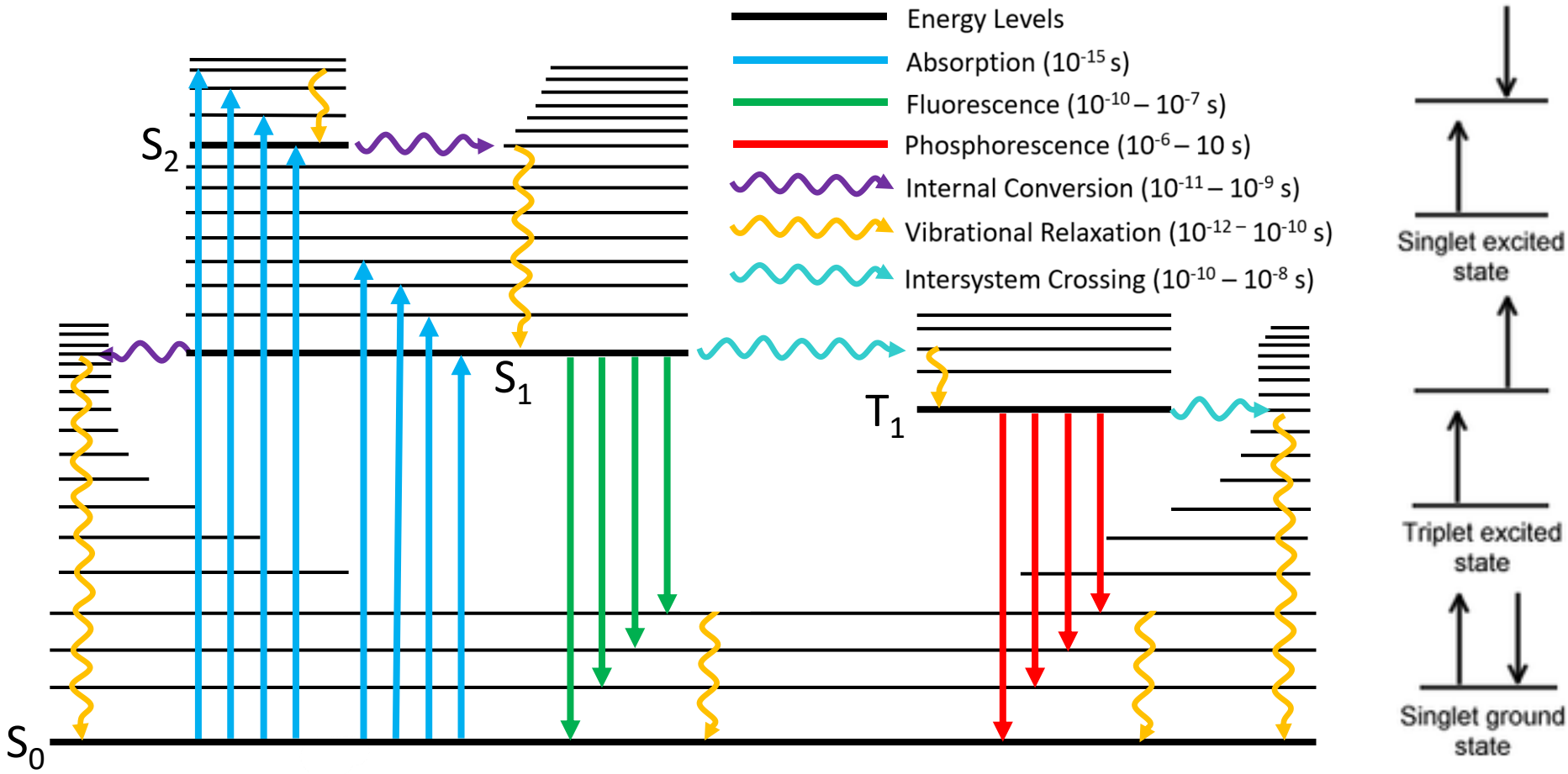
Chem 0010/11

Fluorescence

Dr Sam Cassidy
EngD, MRes, MSc, BSc, FHEA, MRSC, MInstP



Jablonski Diagram



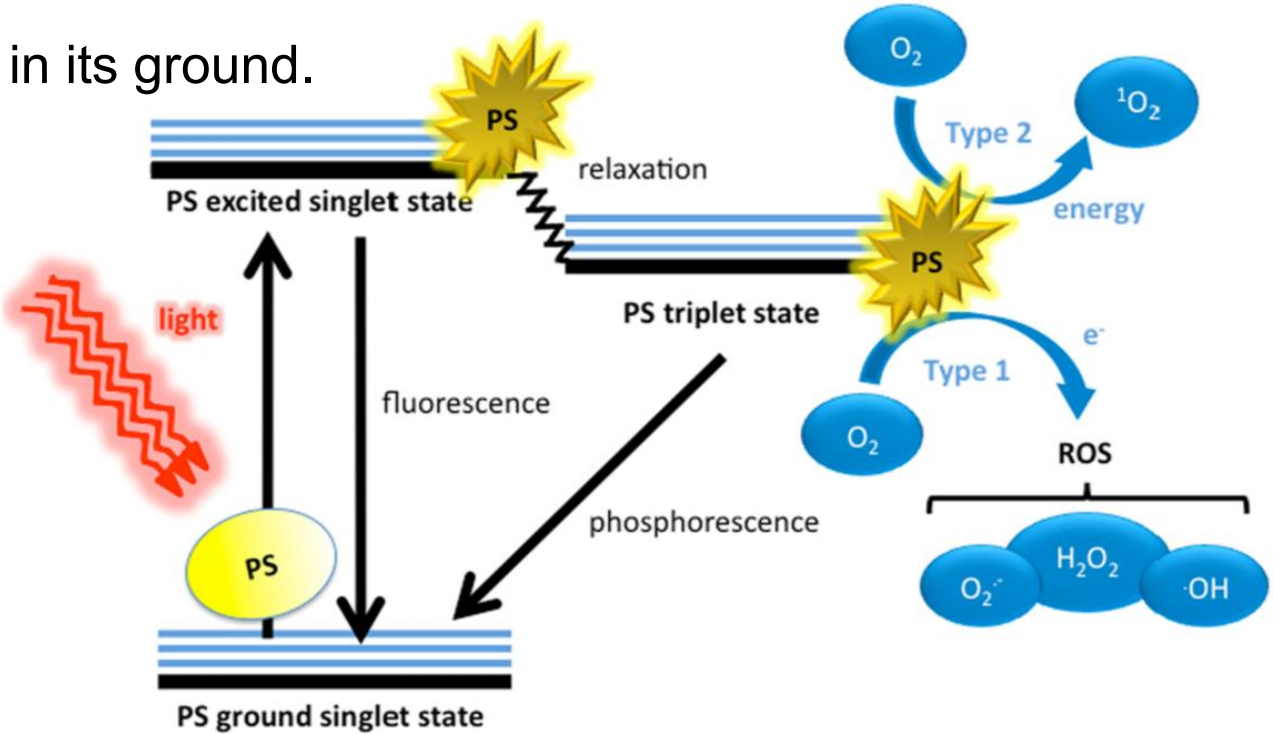
Relaxation

There are a number of ways an excited electron can relax back to a ground state.

1. Radiative decay ($\mathbf{S_1 – S_0}$, $\mathbf{S_2 – S_0}$): Fluorescence, 10^{-9} s.
2. Non-radiative deactivation ($\mathbf{S_1 – S_0}$, $\mathbf{T_1 – S_0}$): Vibrational relaxation, through kinetic energy.
3. Non-radiative deactivation ($\mathbf{S_1 – T_1}$): Intersystem crossing, requires a spin change.
4. Radiative decay ($\mathbf{T_1 – S_0}$): Phosphorescence, 10^{-3} – 10^{2} s.

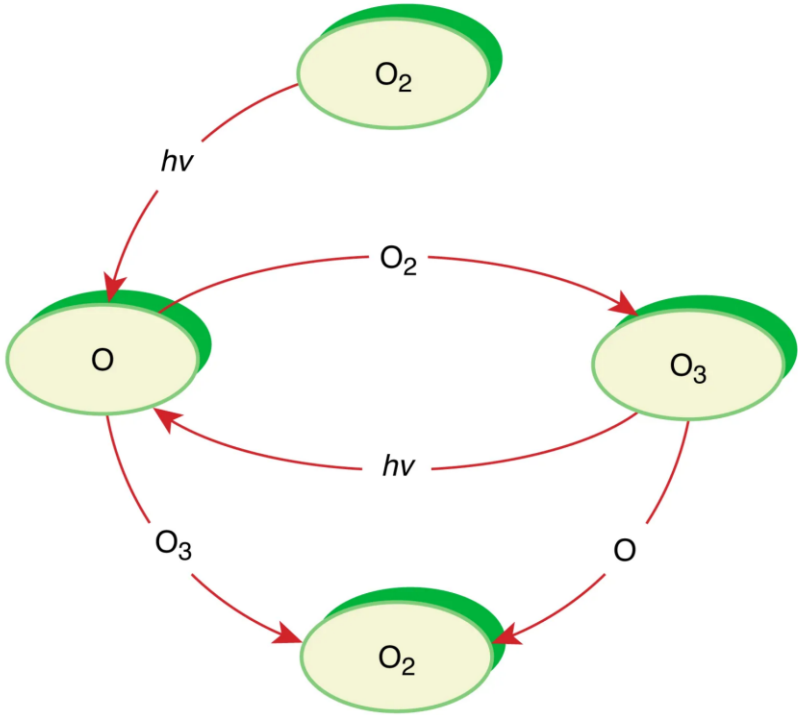
Relaxation

5. Photoreactive channel (**S_1 – Photoproduct, T_1 – Photoproduct**): A molecule in its excited state can exhibit different chemical properties to one in its ground.

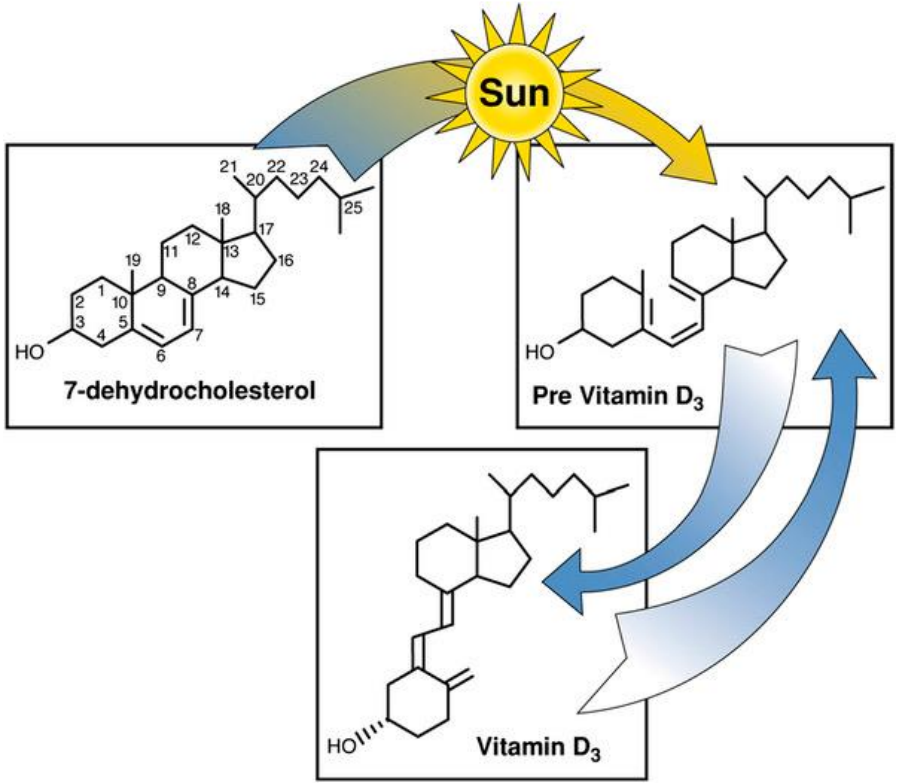


Photochemistry in Biology

Photodissociation

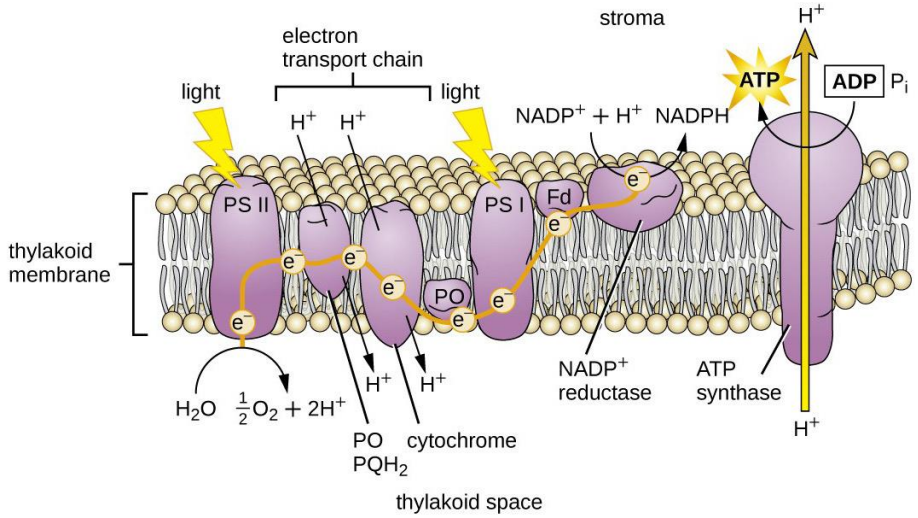


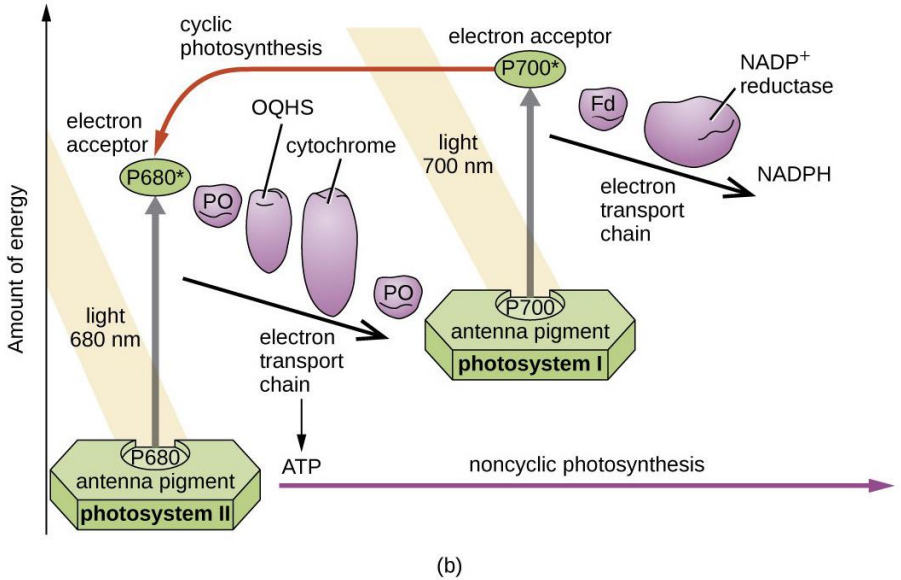
Photorearrangement



Photochemistry in Biology

Photosynthesis



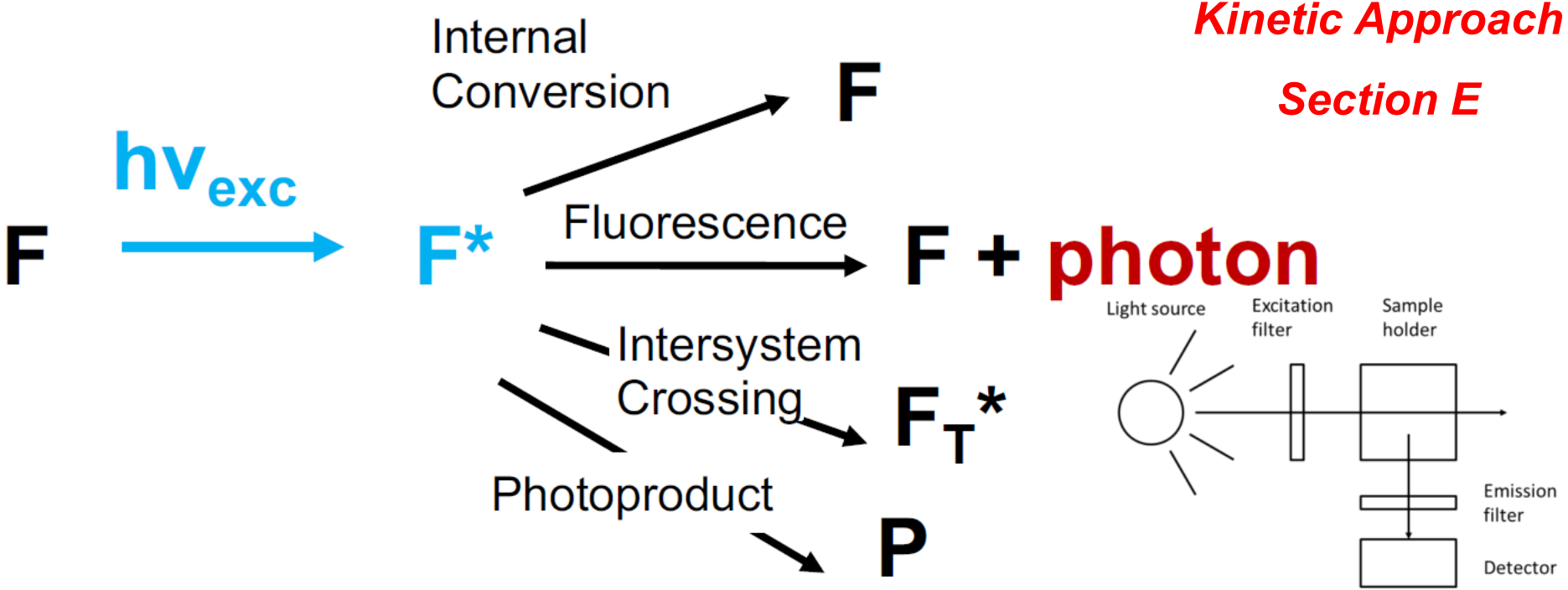


Fluorescence Quantum Yield

$$\phi_f = \frac{I_f}{I_{abs}} = \frac{number\ of\ photons\ emitted}{number\ of\ photons\ absorbed} = \frac{k_f}{k_f + k_{ISC} + k_{IC} + k_P}$$

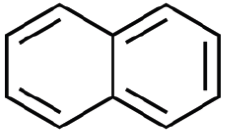
Kinetic Approach

Section E



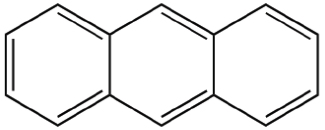
Fluorescence Quantum Yield

The most intense fluorescence is found in compounds containing aromatic groups. The quantum efficiency increases as the number of rings increases.



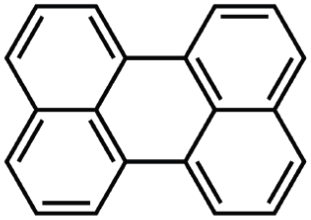
Naphthalene

$\phi_f = 0.19$



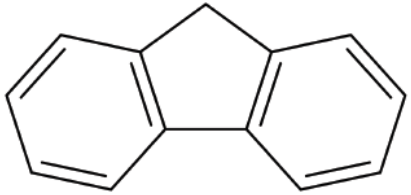
Anthracene

$\phi_f = 0.30$



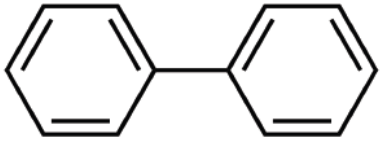
Perylene

$\phi_f = 0.78$



Fluorene

$\phi_f = 1.0$

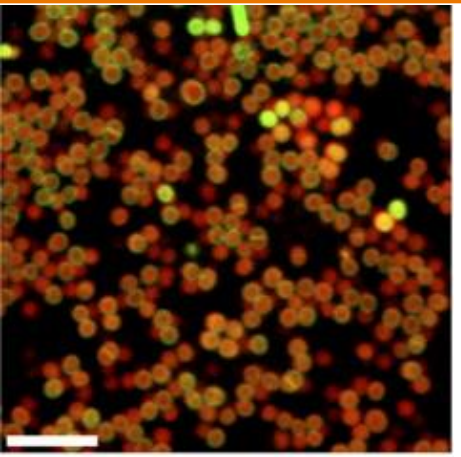


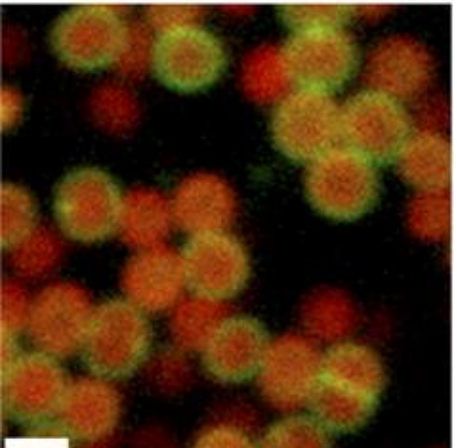
Biphenyl

$\phi_f = 0.2$

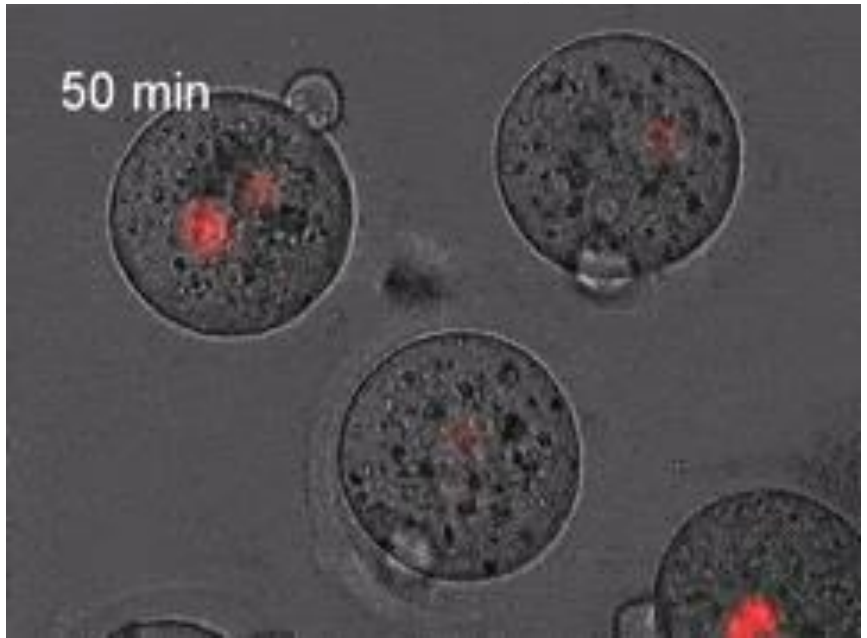
Fluorescence is favoured in molecules with rigid structures. Non-rigid molecules can undergo low-frequency vibration which accounts for small energy loss leading to enhanced internal conversion.

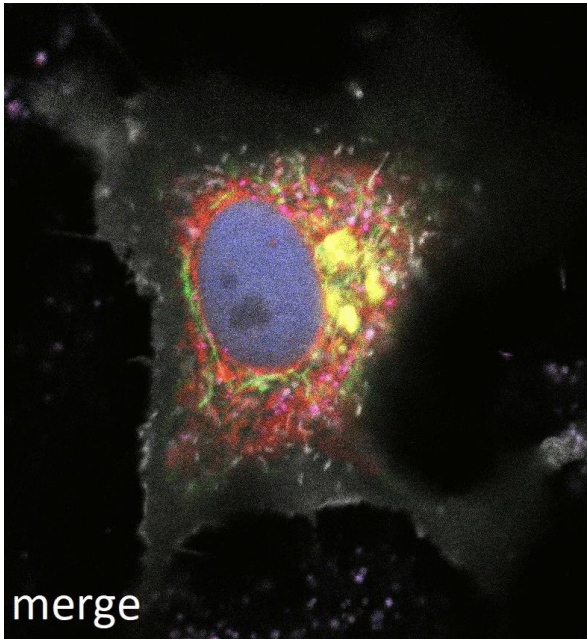
Fluorescent Imaging





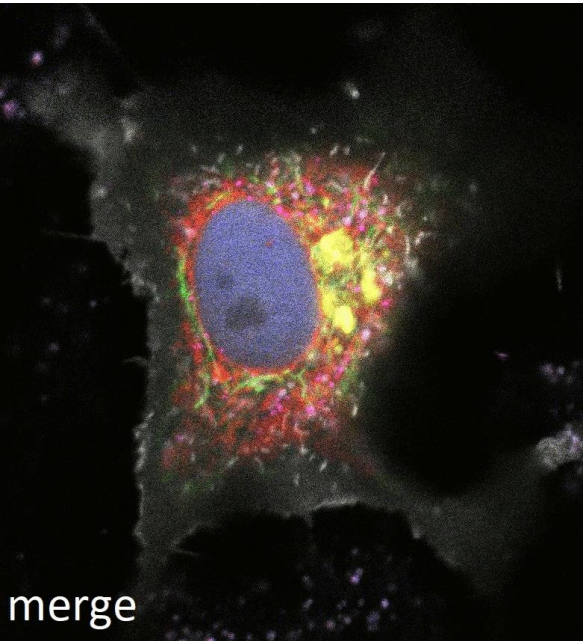
Can we go “Beyond the Diffraction Limit”?

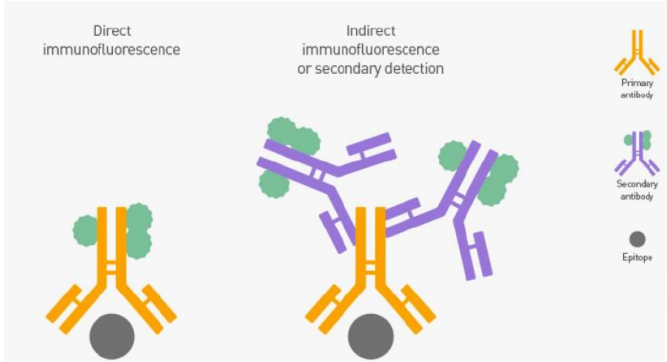


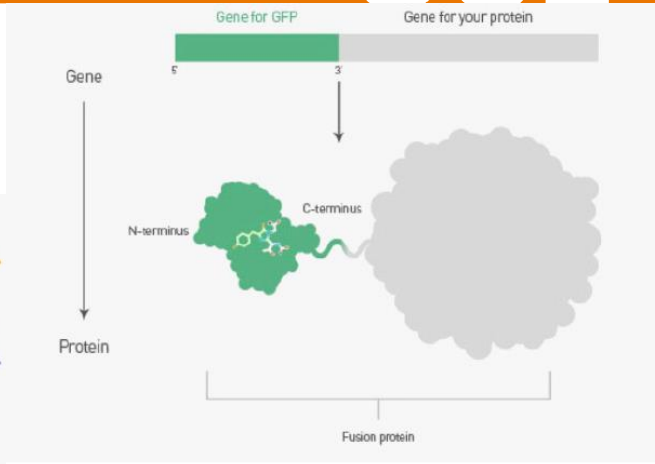


Fluorescent Imaging

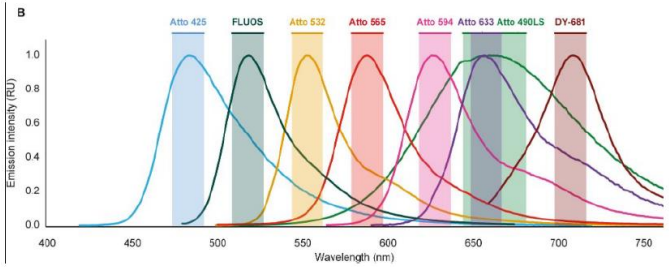
Fluorescent dyes, as well as methods for targeting specific components, have been developed.

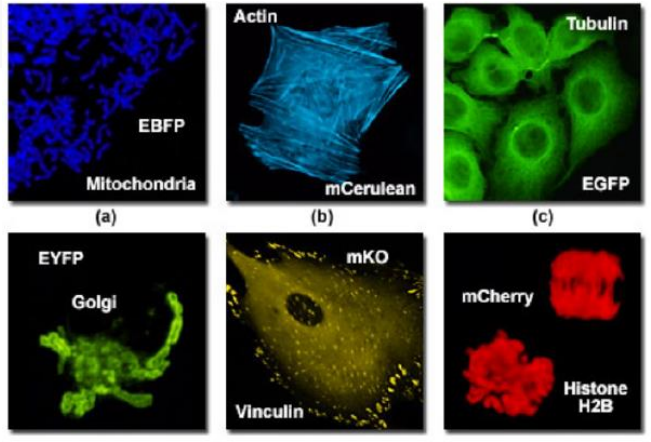




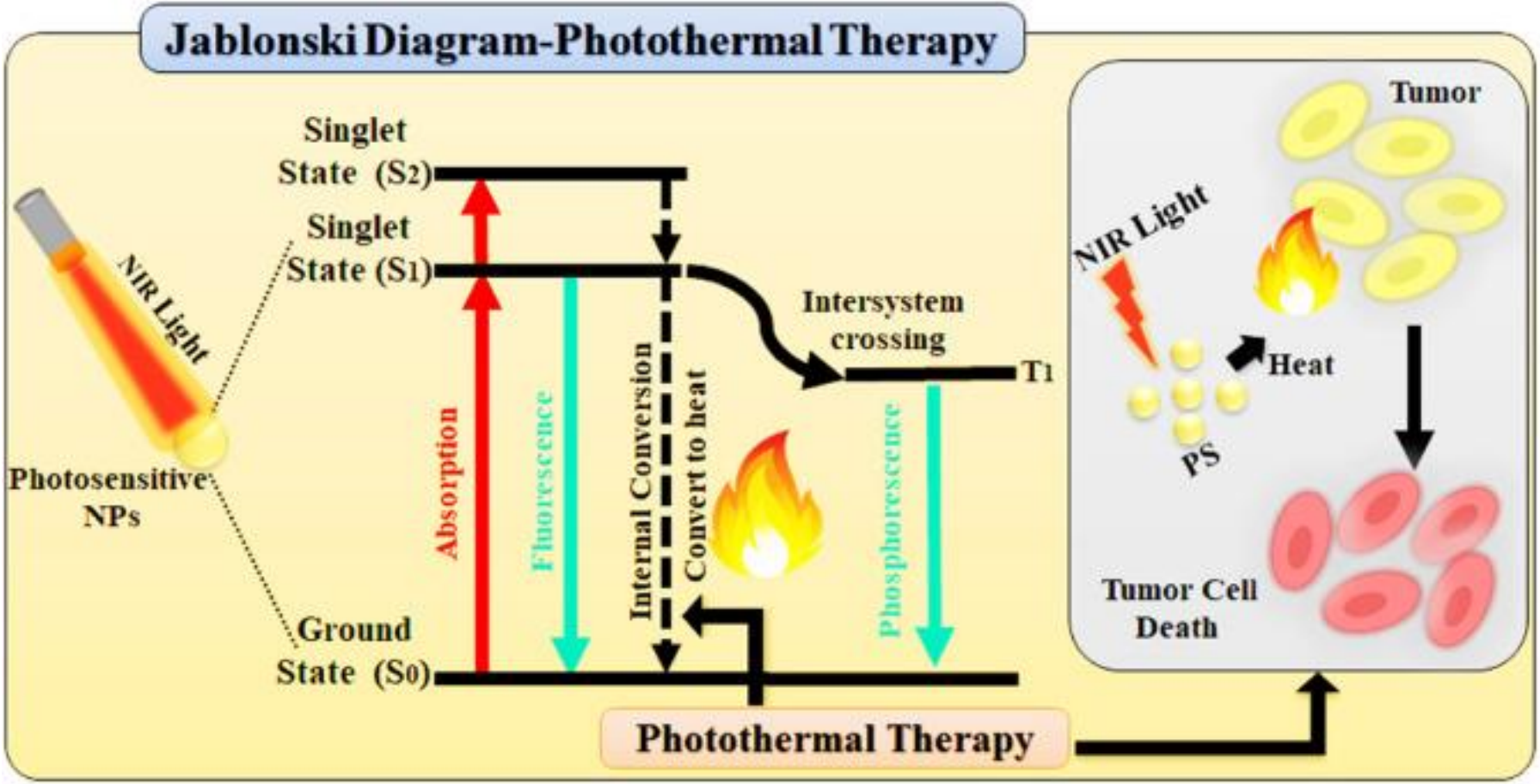


Multiple colors using different species





Photothermal therapy



Summary

- Understand the Jablonski diagram
- Understand the different pathways of relaxation
- Fluorescence tagging
- Fluorescence microscopy
- Cancer treatment